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**Taskio**

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**The candidate confirms that the work submitted is their own and appropriate credit has  
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# CERTIFICATE OF APPROVAL

It is to certify that the final year project of BS (CS) “**Taskio**” was developed by **Muhammad Abdullah (CIIT/SP22-BCS-061)** and **Muhammad Qasim Omer (CIIT/SP22-BCS-085)** and **Aqib Anjum (CIIT/FA21-BCS-170)** under the supervision of “Ms. Raheela Shahzadi ” and co supervisor “CO-SUPERVISOR NAME” and that in (their/his/her) opinion; it is fully adequate, in scope and quality for the degree of Bachelors of Science in Computer Sciences.

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# Executive Summary

The emergence of the gig economy has revolutionized traditional employment structures, leading to a substantial increase in digital freelancing platforms worldwide. While conventional platforms address the basic need for connecting clients (posters) and service providers (freelancers), they often suffer from significant inefficiencies related to matching accuracy and fair pricing transparency. Users frequently face challenges such as wading through irrelevant job listings, encountering fraudulent posts, or struggling to determine competitive rates, thereby necessitating a more intelligent and streamlined solution. The Taskio system is developed as an AI-powered marketplace designed to overcome these prevalent drawbacks. It operates as a full-stack platform built on Django and Django REST Framework (DRF), utilizing machine learning models to enhance user experience and marketplace integrity. The core innovation of Taskio lies in its implementation of a hybrid recommendation system for superior freelancer-to-task matching and a price prediction feature that aids both clients and freelancers in establishing equitable compensation. Furthermore, the platform integrates robust real-time communication tools, secured authentication protocols for reliable login and signup, and asynchronous task processing using Celery and Redis to ensure performance and reliability in task order management and resource (stock) allocation. By emphasizing AI-driven efficiency, security, and user-centric design, Taskio aims to provide a reliable, transparent, and high-performing digital platform for the modern gig workforce.

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Muhammad Abdullah

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# Abbreviations

The abbreviations used throughout this document are defined as follows:

| Abbreviation | Definition                                   |
|--------------|----------------------------------------------|
| <b>AI</b>    | Artificial Intelligence                      |
| <b>API</b>   | Application Programming Interface            |
| <b>BCS</b>   | Bachelor of Science in Computer Science      |
| <b>CI/CD</b> | Continuous Integration/Continuous Deployment |
| <b>DRF</b>   | Django REST Framework                        |
| <b>FR</b>    | Functional Requirement                       |
| <b>ML</b>    | Machine Learning                             |
| <b>NFR</b>   | Non-Functional Requirement                   |
| <b>PC</b>    | Personal Computer                            |
| <b>SDLC</b>  | Software Development Life Cycle              |
| <b>SRS</b>   | Software Requirement Specification           |
| <b>WS</b>    | WebSocket                                    |

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# **Chapter 1**

## **Introduction**

# **1 Introduction**

This initial chapter provides a comprehensive overview of the Taskio project, outlining its context, significance, and the methodological foundation upon which it was built. The subsequent sections detail the essence of the work, its alignment with academic coursework, the background motivation, an extensive review of existing literature, and the rationale for the software engineering approach adopted.

## **1.1 Brief**

Taskio is a cutting-edge, AI-powered digital freelancing marketplace designed to efficiently connect clients seeking services with skilled freelancers offering their expertise, explicitly targeting the local market in Pakistan. The platform's primary outcome is a fully functional, responsive web application featuring sophisticated mechanisms for AI-driven task-worker matching and dynamic price prediction. The entire system is built using the Django Web Framework and the Django REST Framework to provide robust backend APIs for data handling. Key supporting technologies include Celery and Redis, which facilitate asynchronous processing for performance-intensive AI model calls and real-time features like notifications. The methodology chosen is the Iterative Model within an Agile framework, allowing for continuous refinement of core user workflows (like secure login and signup) and incremental integration of complex features (like task order creation and management, and the visualization of the available talent stock (workers). This design choice promotes flexibility, code reusability, and efficient scalability in preparation for future growth.

## **1.2 Relevance to Course Modules**

The Taskio project serves as a capstone application, integrating and demonstrating proficiency across several mandatory courses within the Bachelor of Science in Computer Science curriculum. The system's modular, object-oriented structure directly applies principles from Object-Oriented Programming and Object-Oriented Software Development Methodology. The database schema, transactional integrity, and optimized data access are rooted in knowledge from Database Management Systems and the application of SQL queries. The entire web interface, responsive design, and frontend interactivity reflect competencies gained in Web Engineering, utilizing

modern stacks like HTML5, CSS3, and JavaScript. Crucially, the platform's innovative features AI-Powered Matching, Dynamic Pricing, and Sentiment Analysis are direct applications of theoretical and practical skills acquired in Artificial Intelligence and Machine Learning courses. Finally, the deployment strategy, use of containers (Docker/Kubernetes), security hardening, and rigorous testing directly align with the core concepts of Software Engineering, DevOps, and Information Security.

## **1.3 Project Background**

The foundational concept for Taskio emerged from a rigorous analysis of market needs and existing service gaps within the local Pakistani context. Traditional job boards and international freelancing sites, while functional, present significant hurdles: they are often not localized for the socio-economic dynamics of the region, lack support for local payment methods like JazzCash and EasyPaisa, and require complex verification processes. This creates a twofold problem: for clients, finding a reliable, verified professional, especially for non-technical, physical tasks (like Electricians, Plumbers, and Carpenters) is a time-consuming and trust-intensive endeavor, for skilled workers, the lack of a centralized digital presence contributes directly to underemployment and unemployment issues. Taskio's development is explicitly motivated by the need to create a trusted, one-stop, multi-professional digital platform that is optimized for local citizens and civilians. By incorporating AI-driven intelligence from the outset, the project aims to solve the core challenges of inefficient discovery, compensation ambiguity, and low trust, facilitating connections with "just a CLICK"

## **1.4 Literature Review**

The intellectual foundation of the Taskio project is established through a review of scientific literature focusing on digital labor platforms, intelligent matching systems, and predictive modeling, adhering strictly to IEEE citation standards.

The rise of the global gig economy and digital freelancing marketplaces is documented as a major technological and societal shift, with projections showing substantial workforce growth in regions like India. This expansion is heavily reliant on platform technologies that streamline complex operations. Such platforms, however, often face operational challenges related to inefficient data processing and bottlenecks, requiring sophisticated design optimization of data pipelines to

maintain efficiency and service quality. The Taskio architecture directly addresses this by adopting a microservice approach with asynchronous processing components like Celery/Redis to manage high throughput tasks.

A critical area of research centers on overcoming sub-optimal matching in large-scale recruitment and task-alignment systems. Traditional methods, reliant on simple keyword-based relevance, are inadequate for capturing the nuanced skills and experience needed for complex projects. Recent studies propose novel neural retriever architectures and semantic matching techniques to efficiently align project descriptions with freelancer profiles, particularly in multilingual and varied skill environments. Similarly, work on enhancing freelancer matching advocates for hybrid recommendation strategies that integrate Content-Based Filtering (based on skills and task history) and Collaborative Filtering (based on successful past collaborations) to optimize the selection process. The Taskio system integrates these findings through its AI Task-to-Freelancer Matching Algorithm, leveraging semantic and performance weights to calculate compatibility.

In the domain of pricing transparency and prediction, research demonstrates the viability of applying Machine Learning techniques to forecast values in dynamic markets. While often applied to stock markets, methodologies involving deep learning models like Long Short-Term Memory networks or classic regression models (such as Lasso and Ridge Regression) can be adapted to predict service prices based on multiple features including task complexity, worker experience, and market trends. The implementation of Taskio's Dynamic Pricing Model is directly informed by this research, aiming to establish fair and optimal compensation ranges for both the client and the worker, thereby mitigating pricing disputes and increasing platform trust.

The development of trust and quality control in online marketplaces heavily relies on reputation scoring and sentiment analysis. Studies in opinion mining emphasize the ability of Machine Learning and Deep Learning models to process user reviews and feedback, extracting objective measures of customer sentiment. This depth of analysis moves beyond simplistic star ratings, providing the platform with a crucial tool for quality assurance and identifying service strengths.

Finally, the architectural choices for Taskio, rooted in the Django Web Framework, Django REST Framework, and the extensive use of asynchronous processing tools like Celery and the use of WebSockets for real-time communication, are validated by contemporary software engineering practices emphasizing scalable, modular, and high-performance applications. The use of object-

oriented design principles ensures code reusability and maintainability, facilitating future enhancements. The feasibility of integrating various payment APIs and managing user identities further supports the project's practicality and adherence to modern security standards. Comprehensive implementation requires extensive testing, covering both automated and manual procedures to ensure all functional and non-functional requirements are met.

## **1.5 Analysis from Literature Review**

The literature review unequivocally validates the need for a system like Taskio, specifically emphasizing that its proposed AI-enhanced features are critical differentiators in the crowded digital service marketplace.

The fundamental challenge identified is the gap between simple transactional platforms and truly intelligent marketplaces. Taskio's response, the integration of the AI Engine, aligns precisely with the research favoring sophisticated semantic and hybrid matching algorithms over elementary filters. For instance, a basic platform only checks if a "Plumber" is listed; Taskio uses embeddings derived from the task description and the worker's historical profile and reviews (including implicit sentiment scores) to generate a high-fidelity compatibility score. This ensures a higher success rate for the posted "order" and optimizes the "stock" (freelancer talent pool) allocation.

Furthermore, the integration of the Dynamic Pricing Model directly addresses the established issue of opaque pricing, leveraging known regression methodologies adapted from financial forecasting to provide equitable price bands. This feature serves as a transparency tool, increasing user trust and reducing the negotiation friction prevalent in current systems.

Technological choices reinforce the system's viability. The decision to use Django/DRF provides a strong foundation for building the microservice components Matching Service, Payment Service, and User Management Service as recommended for scalable, high-velocity platforms in the gig economy. Using asynchronous task processing (Celery/Redis) ensures that resource intensive Machine Learning calculations do not impair the platform's real-time responsiveness, maintaining seamless user experience during core actions like login, signup, and task browsing. Finally, the platform's localization for Pakistan, including support for payment APIs like JazzCash and EasyPaisa, strategically capitalizes on the identified market gap left by non-localized international

competitors. This grounded approach, supported by robust academic precedents, positions Taskio as a technically sound and market-relevant solution.

## 1.6 Methodology and Software Lifecycle for This Project

The development of the Taskio system was executed using Agile Methodology in conjunction with the Iterative Model. This combined approach was selected to harness the high velocity, adaptive nature of Agile while mitigating the inherent risks and complexities introduced by developing novel AI features within a transactional marketplace.

The Iterative Model defined the major phases (iterations) of the project:

- **First Iteration (Core Foundation):** Focused on basic user workflows, including client and worker registration (signup), authentication (login), profile management, and initial service listing stubs. This delivered a Minimal Viable Product of the platform's shell.
- **Second Iteration (Core Transactional Logic):** Focused on developing the core business logic, including task posting and bidding modules (Quote Submission), and integrating payment gateways in sandbox mode. The first version of the matching engine was implemented.
- **Subsequent Iterations (AI Integration and Enhancement):** Focused on integrating the high-risk, AI-driven components, such as the LightGBM-based ranking system for provider suggestions and the Sentiment Analyzer for post-job feedback. This iterative approach allowed for continuous testing and tuning of the complex Machine Learning models against live data and user feedback before full deployment.



Figure 1.1: Life Cycle Diagram of Agile Methodology

### 1.6.1 Rationale behind Selected Methodology

The combined Agile and Iterative approach provided several critical benefits to the Taskio project development:

1. **Risk Management (AI):** The explicit segmentation of high-risk components (AI Matching/Pricing) into later iterations allowed the team to validate the fundamental transactional platform structure before committing significant resources to uncertain ML development. This minimized the chance of "analysis paralysis" and ensured that the system remained adaptable to technical challenges.
2. **Stakeholder Collaboration and Feedback:** Agile principles ensured constant interaction with stakeholders (potential users, supervisors) through regular sprints and reviews. This continuous feedback loop was vital for localizing the UI/UX effectively for the Pakistani market and validating that the core features met regional requirements.
3. **Scalability and Modularity:** The iterative design naturally promoted the development of a decoupled microservice architecture. Each service (Matching, Payment, User Management) was treated as a separate, testable module, enhancing code reusability and simplifying maintenance and future feature additions.

# **Chapter 2**

## **Problem Definition**

## **2 Problem Definition**

### **2.1 Problem Statement**

The core problem Taskio addresses is the systemic inefficiency and lack of trust prevalent in the local Pakistani digital service market, which can be synthesized into three critical, interconnected issues.

#### **2.1.1 Ineffective Task-to-Worker Matching**

Existing national and international platforms fail to provide high-fidelity matchmaking for service professionals. While international sites offer vast pools of white-collar professionals, they are not optimized for the local, technical, and non-technical workers required for day-to-day life tasks (e.g., Electricians, Masons, Gardeners). The simple keyword matched by generic platforms results in clients shifting through large numbers of irrelevant or unqualified applicants, wasting time and effort for both parties. This suboptimal process directly limits the efficient allocation of the human resource "stock" available on the platform.

#### **2.1.2 Lack of Pricing Transparency and Trust**

A pervasive issue is the ambiguity surrounding fair pricing. Clients struggle to set appropriate budgets for their tasks (orders), leading either to overpaying or setting uncompetitive prices that discourage skilled applicants. Concurrently, workers often lack market insight to accurately quote their services. This price ambiguity, compounded by a lack of central regulation, severely diminishes the trust factor in transactions. The problem is intensified by inadequate verification mechanisms and reputation scoring, further eroding confidence in hiring non-verified service providers for in-person or high-value tasks.

#### **2.1.3 The Socio-Economic Gap and Lack of Localization**

A critical socio-economic problem in Pakistan is the rising rate of unemployment and poverty, particularly affecting skilled daily-wage workers who are often unaware of local digital opportunities. The absence of a centralized, localized digital marketplace that supports simple, accessible workflows and regional payment methods (like JazzCash and EasyPaisa) excludes a large segment of the potential workforce. This gap perpetuates underemployment and prevents the

digital empowerment of the local talent pool, necessitating a platform that is highly localized and simple to use.

## 2.2 Deliverables and Development Requirements

The successful completion of the Taskio project is defined by the fulfillment of all functional and non-functional requirements and the delivery of the following key artifacts, which collectively form the comprehensive system.

### 2.2.1 Core Web Platform (The Marketplace)

The primary deliverable is a responsive, web-based marketplace application. This includes all required user-facing templates (e.g., `templates/index.html`, `templates/dashboard.html`, `templates/post_task.html`) and the core backend API structure built with Django REST Framework.

### 2.2.2 Modular Service Applications:

The system is logically partitioned into modular Django applications, supporting the microservice architecture (Section 4.1), each with dedicated responsibilities:

- **User Management & Auth (apps/users):** Handles secure login, signup, profile creation, and identity verification.
- **Task Management (apps/tasks):** Manages the entire lifecycle of a task (order), including posting, browsing, updates, and status changes.
- **Application Management (apps/applications):** Manages worker quotes and the acceptance/rejection flow.
- **Payment Integration (apps/payments):** Manages secure, logged payments via external APIs.
- **Messaging (apps/messaging):** Provides the real-time, WebSocket-based communication channel.

### 2.2.3 The AI Engine Microservices

Crucial deliverables are the integrated AI modules that drive the "smart" functionality of the platform:

- **AI Matching Module:** The implementation of the AI-Powered Intelligent Matching Algorithm.
- **Dynamic Pricing Service:** The implementation of the Dynamic Pricing Model.
- **Sentiment Analysis Module:** The component that processes text reviews to assess service quality.

## 2.2.4 Documentation and Code Artifacts

The final delivery includes all source code, configuration files and comprehensive documentation. This includes the final project report, the original Software Requirements Specification (SRS), and the Software Design Description (SDD).

## 2.3 Current System

A comparison with prevailing platforms with both global and implicit local systems underscore the necessity and competitive advantage of Taskio.

**Table 2.1: Problems with Existing Systems**

| Application Name  | Weaknesses in Local Context                                                                        | Taskio Proposed Solution                                                                                                     |
|-------------------|----------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>Fiverr</b>     | International-only focus; not localized for Pakistan; high commission fees for local transactions. | Targets local users with region-specific pricing, UI/UX simplicity, and local payment methods (JazzCash, EasyPaisa)          |
| <b>TaskRabbit</b> | Only operational in US/UK; does not support online (remote) tasks; no presence in Pakistan.        | Supports both physical and online tasks for flexibility in the local market.                                                 |
| <b>Upwork</b>     | Complex verification and high barrier to entry; opaque and complicated bidding process.            | Offers simplified application and AI-driven worker matching, enhances trust via basic identity checks and sentiment scoring. |

|                                                      |                                                                                                                                                     |                                                                                                                                                                             |
|------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Implicit Local (Newspaper/<br/>Word-of-Mouth)</b> | Complete lack of verification or reputation score; time-consuming, geographically limited search; no central "stock" inventory of available talent. | Provides a centralized, secure repository of verified profiles (stock), all workers have a persistent, dynamic reputation score based on reviews and AI sentiment analysis. |
|------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

The critical differentiating factor is Taskio's ability to fuse localization with intelligence. Existing platforms either lack local support or lack the intelligent layer to make matching efficient. For instance, AI Matchmaking moves beyond the rudimentary filters of platforms like Upwork by incorporating a worker's psychological profile (derived from sentiment analysis) alongside technical skills, a capability that dramatically improves match quality and trust. This intelligent layer, combined with native support for local payment methods, establishes Taskio as a unique and essential digital utility in the Pakistani market. The platform transforms the chaotic process of finding a worker into a structured, reliable, and automated digital experience.

# **Chapter 3**

## **REQUIREMENT ANALYSIS**

## 3 REQUIREMENT ANALYSIS

This chapter translates the generalized problem statement into a precise set of user goals and definitive system capabilities, culminating in the complete Software Requirements Specification (SRS) for the Taskio platform. This analysis is based on techniques including the Use Case Diagram and is traceable throughout the development lifecycle.

### 3.1 Use Cases Diagram

The entire scope of the Taskio system is visually captured in the Use Case Diagram below. The diagram identifies the primary actors: Client/Consumer, Worker/Tasker, and Admin (Controller), and maps their required interactions with the platform.

**1. Overview** The Use Case Diagram illustrates the functional requirements of the Task Marketplace System and the interactions between the system's external entities (Actors) and its internal functions (Use Cases). The system acts as an intermediary platform connecting service providers with service seekers, overseen by an administrator.

**2. System Boundary** The solid rectangular box labeled "Task Marketplace System" represents the system boundary. It defines the scope of the software; all functionality inside the rectangle is part of the application being developed, while the entities outside (Actors) interact with the system to achieve specific goals.

**3. Actors** The diagram identifies three primary actors interacting with the system:

- **Worker Tasker (Primary Actor - Left):** Represents the service provider who uses the platform to find work, submit quotes, and manage their schedule.
- **Client Consumer (Primary Actor - Left):** Represents the end-user who posts jobs, reviews offers and hires workers for specific tasks.
- **Admin Controller (Secondary Actor - Right):** Represents the system administrator responsible for maintaining platform integrity, verifying users, and overseeing task management.

**4. Use Cases and Associations** The system functionality is broken down into specific Use Cases, linked to actors via solid association lines:

**A. Common Functionality**

- **Login or Signup:** All three actors (Worker, Client, and Admin) utilize this use case to authenticate and access the system securely.
- **Rating and Review:** Both Workers and Clients can access this feature to leave feedback, ensuring quality assurance and trust within the marketplace.
- **Voice Assistance:** Both Workers and Clients interact with the system via voice commands for accessibility and ease of use.
- **My Tasks:** A central dashboard feature used by both Workers and Clients to track ongoing, pending, or completed jobs.

**B. Worker Tasker Specific Functionality**

- **Profile Verification and Credential Validation:** Workers submit their credentials for validation to ensure legitimacy.
- **Submit or Receive Quotes:** Workers use this to bid on available tasks or receive counter offers.
- **Automated Scheduling:** Allows workers to manage their availability and organize accepted jobs efficiently.

**C. Client Consumer Specific Functionality**

- **Post Task or Create Job:** The core function for Clients to initiate a request for service.
- **Review Offers:** Clients browse the quotes submitted by workers for their specific posts.
- **Hire Tasker:** The finalization step where a client selects a Worker for a job.

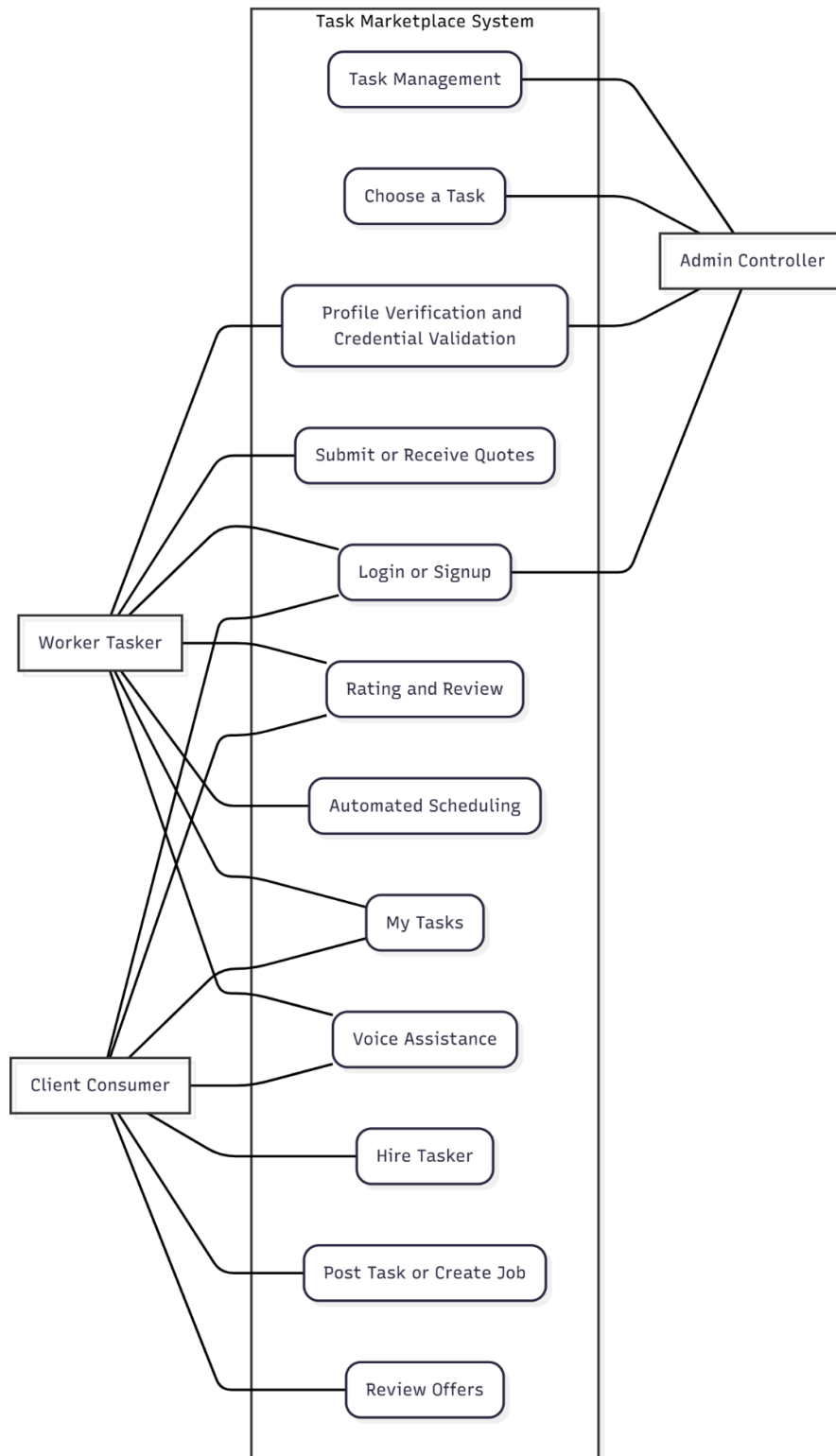
**D. Admin Controller Specific Functionality**

- **Profile Verification and Credential Validation:** The Admin oversees the validation process to approve or reject worker credentials.

- **Task Management:** The admin has high-level control over tasks posted on the platform (e.g., removing inappropriate content).
- **Choose a Task:** Represents the Admin's ability to select specific tasks for detailed review or moderation.

**5. Summary of Interactions** The diagram emphasizes a separation of concerns where the **Admin** (right side) focuses on governance and verification, while the **Worker** and **Client** (left side) focus on the transactional aspects of the marketplace—posting, bidding, and completing tasks.

The diagram illustrates the critical user activities, from authentication (Login/Signup) to task lifecycle management (Post Task/Create Job, Task Management) and the quality control loop (Rating and Review).



**Figure 3.2: Use Case Diagram of Taskio**

The Use Case Diagram in Figure 3.1 organizes the system requirements into logical blocks. The Client focuses on commissioning work (Post Task), selecting candidates (Review Offers, Hire Tasker), and feedback (Rating and Review). The Worker is focused on finding jobs (Choose a Task), applying (Submit/Receive Quotes), and managing their service status (Automated Scheduling). The admin acts as the controller, managing tasks, user accounts, and verification processes to maintain platform integrity.

## 3.2 Detailed Use Case

A detailed description of critical use cases ensures clarity on interaction flows, system responsibilities, and the precise functionality expected of the platform. By expanding the fundamental scenarios into step-by-step processes, this section provides the granular detail necessary for implementation, tracing the system's behavior against the core functional requirements (FRs) defined in the Software Requirements Specification (SRS).

### 3.2.1 Detailed Use Case: Secure User Registration

This case outlines the process for a new user to securely establish an account, defining their primary role on the platform and ensuring the integrity of the user base.

**Table 3.2: Secure User Registration (Sign Up)**

| Attribute             | Description                                                                                                                                                |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case ID</b>    | FR-01 (User Registration and Authentication)                                                                                                               |
| <b>Use Case Name</b>  | User Registration                                                                                                                                          |
| <b>Actors</b>         | Client/Consumer, Worker/Tasker, Admin                                                                                                                      |
| <b>Description</b>    | Users must successfully create and verify an account before accessing application features, including selecting their specialized role (Client or Worker). |
| <b>Trigger</b>        | User attempts to access restricted application features or uses the platform for the first time.                                                           |
| <b>Pre-Conditions</b> | User must open the Taskio web application on a device (PC or mobile). 2. User must have an active internet connection.                                     |
| <b>Normal Flow</b>    | 1. User navigates to the registration page.                                                                                                                |

|                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                          | <p>2. User selects their role (Client or Worker).</p> <p>3. User enters required credentials (name, email, phone, and secure password).</p> <p>4. The Task Controller validates the input data against defined security policies (e.g., password complexity, email uniqueness).</p> <p>5. The system saves the new User record to the database, setting the appropriate role Enum.</p> <p>6. An asynchronous email task is triggered, sending a verification link.</p> <p>7. The system redirects the user to the verification status page.</p> |
| <b>Alternative Flows</b> | <p>A1: Invalid Role Selection: If the user attempts to proceed without selecting a role, the system prevents form submission and prompts for selection.</p> <p>A2: Password Policy Violation: The system provides explicit, granular feedback detailing which password requirement (e.g., minimum 8 characters, special characters) was not met.</p>                                                                                                                                                                                            |
| <b>Exceptions</b>        | <p>E1: Duplicate Email/Phone: The system displays an error informing the user that the credentials already exist, prompting them to login instead. \</p> <p>E2: Email Service Failure: If the asynchronous email verification task (Celery) fails, the system logs the error and allows the user to request a new verification email upon login attempt.</p>                                                                                                                                                                                    |
| <b>Post Conditions</b>   | <p>1. A new User record is created in the database, with a set role and an unverified status.</p> <p>2. A verification email containing a unique token is successfully sent to the user's provided address.</p>                                                                                                                                                                                                                                                                                                                                 |

### 3.2.2 Detailed Use Case: Secure User Authentication

This use case manages the process by which a registered user securely accesses the platform's personalized features, applying role-based access to the correct dashboard.

**Table 3.3: Detailed Use Case: Secure User Authentication (Login)**

| Attribute                | Description                                                                                                                                                                                                                                                                                                                                                                              |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case ID</b>       | FR-02 (Login/Logout Functionality)                                                                                                                                                                                                                                                                                                                                                       |
| <b>Use Case Name</b>     | Login (Logout)                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Actors</b>            | Client/Consumer, Worker/Tasker, Admin                                                                                                                                                                                                                                                                                                                                                    |
| <b>Description</b>       | Allows verified users to securely authenticate with the system and gain access appropriate to their predefined role.                                                                                                                                                                                                                                                                     |
| <b>Trigger</b>           | User attempts to access the dashboard or any protected API endpoint.                                                                                                                                                                                                                                                                                                                     |
| <b>Pre-conditions</b>    | 1. User must have a successfully created and verified account.<br>2. User must be on the Login page.                                                                                                                                                                                                                                                                                     |
| <b>Normal Flow</b>       | 1. User enters verified email or phone and password.<br>2. The WebApp UI sends credentials to the Task Controller.<br>3. The Task Controller validates credentials against the Database (Figure 4.3).<br>4. The system verifies the user's role Enum (Admin, Worker, or Client).<br>5. Access is granted, and the system redirects the user to their corresponding role-based dashboard. |
| <b>Alternative Flows</b> | A1: Forgot Password: If the user clicks the password reset link, an email is sent containing a secure, time-sensitive token, allowing them to reset their credentials securely.<br>A2: Account Suspended: If the user's status is 'Suspended', the system displays a contact message and denies access.                                                                                  |
| <b>Exceptions</b>        | E1: Incorrect Credentials: System denies access and displays a generic security error ("Invalid email or password"), preventing brute-force attack enumeration.<br>E2: Unverified Account: System denies access and redirects to a status page, offering to resend the verification email link.                                                                                          |

|                        |                                                                                                                                                                                              |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Post Conditions</b> | <ol style="list-style-type: none"> <li>1. The user is successfully authenticated and logged into the system.</li> <li>2. The user is viewing the correct role-specific dashboard.</li> </ol> |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 3.2.3 Detailed Use Case: Post Task

This is the central transactional process where a client initiates a project ('order'), leveraging AI services for pricing guidance.

**Table 3.4: Detailed Use Case: Post Task**

| Attribute                | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case ID</b>       | FR-03 (Task Posting)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Use Case Name</b>     | Adding New Tasks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Actors</b>            | Client/Consumer, Admin                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Description</b>       | Allows a client to define and submit a new task listing to the marketplace, including title, detailed description, and budget.                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Trigger</b>           | Client clicks on the "Post a Task" button on their dashboard.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Pre-conditions</b>    | <ol style="list-style-type: none"> <li>1. Client must be logged in.</li> <li>2. Client must have completed all mandatory profile fields, including identity verification.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Normal Flow</b>       | <ol style="list-style-type: none"> <li>1. Client navigates to the 'Post Task' page.</li> <li>2. Client enters Task details.</li> <li>3. Upon input, the Application Layer triggers a non-blocking call to the AI Dynamic Pricing Service.</li> <li>4. The system displays the AI-suggested budget range to the Client.</li> <li>5. Client sets the final budget and submits the form.</li> <li>6. The Task Controller validates data, creates the Task record, and sets status to pending.</li> <li>7. The system queues for an asynchronous Celery Task to run the AI Matching Engine.</li> </ol> |
| <b>Alternative Flows</b> | A1: No Budget Set: If the Client opts for bidding, the budget field is set to zero, and the system proceeds, relying on Workers' quotes to define the market price.                                                                                                                                                                                                                                                                                                                                                                                                                                |

|                        |                                                                                                                                                                                                                                                                                   |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                        | A2: AI Price Model Fails: The system displays a fallback market average price and alerts the Client that the price guidance is based on historical averages only.                                                                                                                 |
| <b>Exceptions</b>      | E1: Invalid Data: If mandatory fields are missing or data is malformed, the system displays an error and prompts the Client to correct the details. E2: AI System Timeout: If the AI call takes too long (exceeds timeout), the system defaults to A2 and logs the timeout error. |
| <b>Post Conditions</b> | 1. A new Task record is stored in the database, with pending status.<br>2. The AI Matching process is initiated in the background, preparing to send notifications to highly qualified Workers.                                                                                   |

### 3.2.4 Detailed Use Case: Quote Submission by Worker

This case details the action a Worker takes to apply for an available task, submitting a financial proposal (quote) for Client review.

**Table 3.5: Detailed Use Case: Quote Submission by Worker**

| Attribute             | Description                                                                                                                                                                                                                                    |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case ID</b>    | FR-05 (Quote Submission by Workers)                                                                                                                                                                                                            |
| <b>Use Case Name</b>  | Quote Submission                                                                                                                                                                                                                               |
| <b>Actors</b>         | Worker/Tasker                                                                                                                                                                                                                                  |
| <b>Description</b>    | Allow a Worker to respond to an available task (order) by proposing a price and indicating their interest in performing the work.                                                                                                              |
| <b>Trigger</b>        | Worker views a task listing and initiates the application process.                                                                                                                                                                             |
| <b>Pre-conditions</b> | 1. Worker must be logged in and have a verified profile.<br>2. The target Task must have a pending status (available for quotes).                                                                                                              |
| <b>Normal Flow</b>    | 1. Worker views the task detail page.<br>2. Worker clicks the 'Submit Quote' button.<br>3. Worker enters the proposed amount and a brief cover message. 4. Worker executes the send quote method.<br>5. The system validates the quote amount. |

|                         |                                                                                                                                                                                                                                                                                                       |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                         | 6. A new Quote record is created and saved, linking the Worker to the Task.<br>7. The Client (Task Poster) receives a real-time notification (via WebSockets) and an email alert that a new quote has been received.                                                                                  |
| <b>Alternative Flow</b> | A1: Already Applied: If the Worker attempts to apply again, the system prompts them to modify their existing quote instead of creating a new one.<br>A2: Task Assigned: If the Task status changes to ACTIVE immediately before submission, the system rejects the quote and displays a notification. |
| <b>Exceptions</b>       | E1: Quote Integrity Violation: If the proposed amount is deemed potentially fraudulent (e.g., significantly below minimum wage floor), the submission is flagged for Admin review before being presented to the Client.                                                                               |
| <b>Post Conditions</b>  | 1. A new Quote record is stored in the database, with quoteStatusEnum set to submitted.<br>2. The Client is successfully notified, and the Worker's Quote is added to the list viewable by the Client.                                                                                                |

### 3.3 Functional Requirements

Functional Requirements define the features and services the system must provide to the user, directly supported by the implementation codebase.

#### 3.3.1 User Registration and Authentication

The platform shall provide a secure and efficient registration process (Client/Worker role selection) with mandatory email verification to ensure basic identity vetting.

- The system shall enforce password complexity requirements during signup, including minimum length (8 characters), special characters, and numeric characters.
- The system shall allow users to securely log in using their verified credentials and provide a 'Forgot Password' flow.

#### 3.3.2 Task Management

The system shall allow logged-in Clients to seamlessly post tasks by providing details like title, detailed description, category, and desired completion deadline.

- The system shall automatically transition the task status through its defined lifecycle.
- The system shall allow users to browse and filter available task listings based on category, location, and price.

### **3.3.3 Quote Submission and Management**

The system shall enable Workers to submit a price quote for any available task, and clients shall be able to view and manage all received quotes.

- The Client shall be able to 'Hire' an applicant, changing the task status to activate and triggering a payment escrow setup.

### **3.3.4 AI-Powered Intelligent Matching**

The system shall employ a hybrid AI algorithm that dynamically calculates a compatibility score between a posted task and available workers.

- The matching algorithm shall consider skill matching, past review ratings, and the Worker's historical performance (completion rate, earnings) to generate the match score.
- The system shall send proactive notifications only to the top-ranked Workers upon a new task posting, optimizing outreach.

### **3.3.5 Dynamic Pricing Model**

The system shall utilize an AI-based regression model to continuously analyze market data and task attributes to recommend an optimal budget range for the Client during task posting.

- The pricing model shall consider factors including task category, complexity (description length/keywords), location, and average worker hourly rates for the region.

### **3.3.6 Sentiment Analysis**

The application shall process and analyze text reviews and feedback through AI-driven sentiment analysis to identify the emotional tone (Positive, Negative, Neutral) of the interaction.

- The resulting sentiment score shall be permanently stored as part of the Worker's reputation profile, contributing to the AI Matching Score.

### **3.3.7 Real-time Communication**

The platform shall provide a persistent, secure, and real-time messaging system (WebSockets) for direct communication between the client and the engaged worker.

- The system shall support AI-powered translation of messages to overcome potential language barriers between users.

### **3.3.8 Identity Verification**

The system shall provide an optional, but highly encouraged, identity verification process for workers, including the submission of a CNIC image.

- The verification module shall use OCR and image recognition techniques to cross-validate the identity document against the Worker's profile information.

### **3.3.9 Payment Integration**

The system shall facilitate secure transactions via integration with regional payment gateways (JazzCash, EasyPaisha) and international methods (Stripe).

- The platform should utilize an escrow system, holding funds securely upon task assignment and releasing them to the worker only upon the Client's approval of the completed task.

## **3.4 Non-Functional Requirements**

Non-Functional Requirements define the quality attributes and constraints of the system, impacting on the overall user experience and technical integrity.

### **3.4.1 Performance**

The system must deliver high responsiveness, especially for user-facing actions (e.g., loading dashboards, searching tasks).

- Latency for displaying task listings should not exceed 500ms under typical load (up to 1,000 concurrent users).
- All AI model predictions (Price, Matching) shall be offloaded to asynchronous workers (Celery) to prevent blocking the main API thread, with results returned via background updates.

### **3.4.2 Scalability**

The system architecture must be horizontally scalable to support future growth in the user base and increased transactional volume.

- The modular, microservice-based architecture shall allow independent scaling of high-demand services (e.g., the AI Matching Microservice and the WebSocket Messaging Service).
- The data layer must support polyglot persistence (SQL for transactional data; MongoDB/NoSQL for profiles/chat logs) to optimize access performance and storage needs.

### **3.4.3 Security and Privacy**

Robust measures must be implemented to secure user data and ensure privacy.

- All authentication processes shall use modern standards with encrypted storage of credentials.
- Data-in-transit (including chat messages and payment details) must be secured using TLS/SSL encryption.
- User access must be strictly controlled by role-based permissions (Client vs. Worker vs. Admin) to prevent unauthorized data access.

### **3.4.4 Usability**

The platform's interface (UI/UX) must be intuitive and optimized for users with varied levels of technical literacy, ensuring accessibility.

- The entire application shall use a responsive design to function seamlessly across desktop, laptop, and Android devices.
- AI-enhanced features (like pricing suggestions) must enhance the experience without introducing undue complexity or confusion.

### **3.4.5 Maintainability**

The codebase must be modular, well-documented, and adhere to clean coding standards to facilitate updates and long-term maintenance.

- All AI models shall be contained within distinct, modular services (AI Matching Microservice, Pricing Service Module) to allow for independent retraining and updating without affecting the core web platform.
- The deployment process must be automated using Continuous Integration/Continuous Deployment (CI/CD) pipelines.

# **Chapter 4**

## **Design and Architecture**

## 4 Design and Architecture

### 4.1 System Architecture

The Taskio system is implemented using a Microservice-Oriented Three-Tier Architecture, which comprises the Presentation Layer, the Application Layer (Business Logic), and the Data Layer. This structure ensures a high degree of modularity and independent scalability, essential for a platform that handles complex transactional flows alongside resource-intensive AI computations.

### 4.2 Architecture Design Diagram

This diagram represents the N-tier architecture of the TaskIO platform, illustrating the separation of concerns into distinct logical layers: Client, Presentation, Application, Messaging, and Data. This modular design ensures scalability, maintainability, and the efficient handling of both synchronous user requests and asynchronous background tasks.

#### Detailed Layer Breakdown:

##### 1. Client Tier (Frontend):

- **Access Point:** The entry point for all interactions is the "Client Tier," which consists of a "Web Browser and Mobile-Responsive UI". This ensures cross-platform accessibility for both desktop and mobile users.

##### 2. Presentation Layer (API & Gateway):

- **Core Framework:** The backend interface is managed by the "Presentation Layer," utilizing "Python Django and WebSockets" to handle incoming HTTP requests and persistent connections.
- **Real-Time Features:** This layer includes a dedicated "Real-time Notification" module powered by "Django Channels," enabling instant updates for users.
- **UX Components:** It also hosts "Accessibility Feature and Responsive Design" modules to ensure interface compliance.

##### 3. Messaging Infrastructure (Asynchronous Processing):

- **Message Broker:** A critical component for performance is the "Messaging Infrastructure," which uses "RabbitMQ and Redis Broker" to manage task queues.
- **Task Execution:** This infrastructure feeds into "Celery and AI Task Workers," allowing heavy computational tasks (like AI price prediction or matching algorithms) to run in the background without blocking the main user interface.

**4. Application Layer (Business Logic Services):** This layer houses the core functional microservices of the platform:

- **Matching Service:** Utilizes "LightGBM-based Ranking" to intelligently pair freelancers with tasks.
- **Payment Service:** Manages financial transactions via "Payment and Billing Service" with "Escrow" functionality for secure holdings.
- **Core Management:** Includes "Task Management Service (CRUD)" for project lifecycle operations and "User Management" for account handling.
- **Security:** A "Verification Service" performs "ID Document Validation" to ensure platform trust.

**5. Data Layer (Polyglot Persistence):** The system employs a "Polyglot" database strategy, optimizing storage based on data type:

- **Relational Data:** An "SQL Server" is used for structured, transactional data such as "Tasks," "Bids," and "Payments".
- **Unstructured Data:** A separate NoSQL-style storage handles semi-structured data including "User Profiles," "Chat Logs," and "Feedback".

#### **Key Architectural Highlights for Your Report:**

- **Decoupling:** The use of a "Messaging Infrastructure" (RabbitMQ/Celery) separates the user interface from heavy backend processing, ensuring the app remains fast even during complex AI calculations.
- **Scalability:** The service-oriented approach in the Application Layer (separating Payments, Matching, and Tasks) allows independent scaling of specific features.

- **Data Integrity vs. Flexibility:** The split between SQL (for money/tasks) and NoSQL (for chats/profiles) optimizes both transactional safety and retrieval speed.

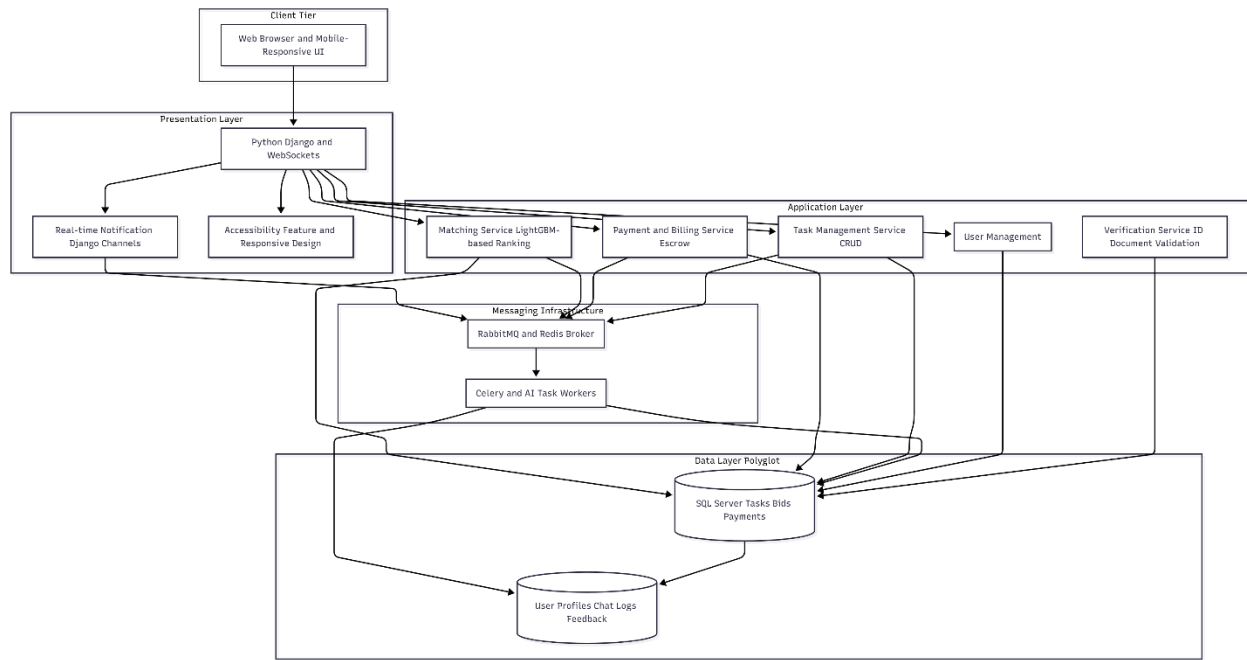


Figure 4.3: Architecture Design Diagram of Taskio

### 4.2.1 Layer Components and Responsibilities

**1. Presentation Layer:** Acts as the client-facing interface. It is built with Python (Django) templates to deliver responsive and accessible UIs. The primary function is to handle user input, display data fetched from the Application Layer, and manage the Real-time Notification system using WebSockets (Django Channels).

**2. Application Layer:** Contains the core business logic, implemented as decoupled microservices.

- **Matching Service:** Employs the LightGBM-based ranking system for intelligent worker matching.
- **Task Management Service:** Manages the lifecycle of a Task, handling all CRUD operations and state transitions.
- **Payment and Billing Service:** Manages financial transactions, Stripe integration, and the internal escrow system.

- **Asynchronous Processing:** This is the critical performance bottleneck solution. It uses RabbitMQ (or Redis) as a message broker and Celery workers to execute resource-intensive tasks like AI matching, sentiment analysis, and automated scheduling.

**3. Data Layer:** Adopts a Polyglot Persistence strategy for data optimization.

- **SQL Server (Transactional Data):** Used for highly structured data requiring high integrity and strict relationships, such as tasks, bids, payments, and audit logs.
- **MongoDB (Semi-structured Data):** Used for flexible data storage, such as detailed user profiles, raw chat logs, and comprehensive job feedback metadata.
- **Redis/RabbitMQ:** Used for catching frequently accessed data (e.g., top-ranked providers) and managing asynchronous workflows

## 4.3 Data Representation

The Data Design focuses on defining the key structural components of the system through a **Class Diagram**, which serves as the blueprint for the relational and non-relational data models, ensuring consistency, integrity, and scalability across all application modules.

### 4.3.1 Class Diagram

The diagram represents the architecture of a Task Marketplace Application. The system connects clients (who need work done) with workers (who perform the work), facilitated by an Administrator and enhanced by an AI Engine.

#### 1. User Hierarchy (Inheritance)

The system uses an inheritance structure to manage different user roles efficiently.

- **User (Parent Class):**
  - This is the base class for all actors in the system.
  - **Attributes:** Holds common data like userId, name, email, password, phone, role (linked to RoleEnum), and address.
  - **Methods:** login(), updateProfile(), logout().

- **Admin (Child Class):**
  - Inherits from User.
  - **Responsibilities:** Managing the platform's integrity.
  - **Methods:** verifyUser() (KYC check), manageTask() (overseeing task content), and banUser() (removing bad actors).
- **Client (Child Class):**
  - Inherits from User. Represents the person hiring help.
  - **Responsibilities:** Creating tasks and hiring.
  - **Methods:** postTask(), viewQuotes() (seeing bids), hireWorker() (selecting a candidate), and approveWork() (finalizing a job).
- **Worker (Child Class):**
  - Inherits from User. Represents the service provider.
  - **Attributes:** Includes specific fields like skills (List), rating (float), and experienceLevel.
  - **Methods:** browseTasks(), sendQuote() (bidding on a task), and submitWork().

## 2. Core Entities

These classes handle the business logic of the transactions.

- **Task:**
  - The central object of the system.
  - **Attributes:** taskId, title, description, budget, location, status (linked to TaskStatusEnum), createdAt, and clientId.
  - **Methods:** updateStatus() and getDetails().
  - **Relationship:** A **Client** can post multiple **Tasks** (1 to 0..\* relationship).

- **Quote:**
  - Represents a bid or proposal sent by a worker for a specific task.
  - **Attributes:** quoteId, taskId, workerId, amount, status, and sentAt.
  - **Methods:** accept() and reject().
  - **Relationship:** A **Task** can have multiple **Quotes**, and a **Worker** can send multiple **Quotes**.
- **Review:**
  - Represents feedback left after a task is completed.
  - **Attributes:** reviewId, taskId, reviewerId, rating, comment, and createdAt.
  - **Relationship:** Linked to a specific **Task** and **Worker**.

### 3. Advanced Functionality

- **AIEngine:**
  - This is a utility class providing intelligent features to the platform.
  - **matchClientToWorker(task):** Algorithms to suggest the best worker for a job.
  - **calculateDynamicPricing(task):** Suggests pricing based on demand/market.
  - **analyzeSentiment(text):** Likely used to monitor reviews or chat messages for safety/quality assurance.
  - **Connections:** The diagram shows the AI Engine interacting with Task, Quote, and Review objects.

### 4. Enumerations (Enums)

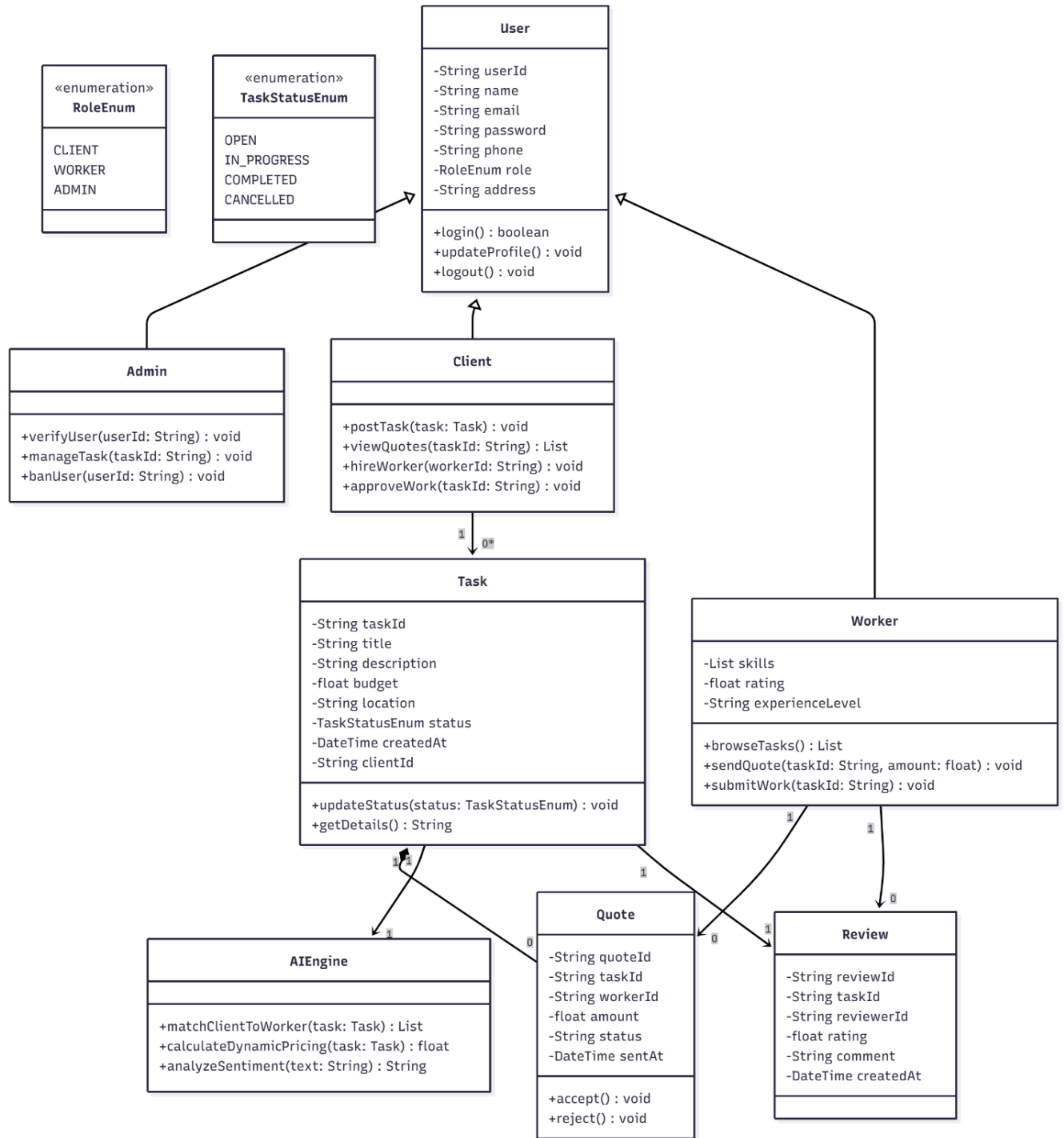
These define fixed sets of constants used across the system to prevent errors.

- **RoleEnum:** Defines the user types: CLIENT, WORKER, ADMIN.
- **TaskStatusEnum:** Defines the lifecycle of a task: OPEN, IN\_PROGRESS, COMPLETED, CANCELLED.

### Summary of Relationships

1. **Client ↔ Task:** One-to-Many. One client posts many tasks.
2. **Worker ↔ Quote:** One-to-Many. One worker sends many quotes.
3. **Task ↔ Quote:** One-to-Many. One task receives many quotes.
4. **Task ↔ Review:** One-to-One (implied). A completed task generates a review.

The following figure represents the static view of the object model, detailing the core entities, their attributes, methods, and relationships.



**Figure 4.4: Class Diagram of Taskio**

The **Class Diagram** provides a foundational, static object-oriented view of the Taskio system's core structure, detailing the fundamental entities, their attributes, and the relationships that govern the entire application.

1. **Core Entities and Inheritance:** The diagram establishes a base class, User, with common attributes like userId, email, and password. This class is then specialized via inheritance into Client (representing the task poster), Worker (representing the service provider), and Admin (for platform management).
2. **Transactional Entities:** Key transactional entities define the marketplace workflow:
  - **Task:** Represents the job post or 'order,' including attributes like title, description, and budget.
  - **Quote:** Represents a worker's application/bid on a task, storing the amount proposed and the status of the quote.
  - **Review:** Captures user feedback with attributes like rating and comment, which are crucial inputs for the AI Engine.
3. **Encapsulation of Intelligence:** The diagram highlights the AI Engine as a dedicated Utility Class. This class encapsulates all the platform's intelligent logic, including the core methods: match client to worker, calculate dynamic pricing, and perform sentiment analysis.
4. **Relationships and Methods:** Relationships clearly define how objects interact (e.g., a client posts a task, a worker sends a quote. Specific methods, such as posting tasks on the Client class and sending quotes on the Worker class, define the allowed actions each role can take, forming the functional core of the system.

### 4.3.2 Data Dictionary Summary

The Class Diagram serves as a high-level data dictionary. Below is a summary of key entity fields:

**Table 4.6:Data Dictionary Summary**

| Entity           | Key Field             | Data Type     | Constraint/Purpose                                                   |
|------------------|-----------------------|---------------|----------------------------------------------------------------------|
| <b>User</b>      | role                  | roleEnum      | Defines access level: Admin, Worker, or Client.                      |
| <b>Worker</b>    | skills                | List<string>  | Used as a primary input for the AI Matching Algorithm.               |
| <b>Task</b>      | budget                | float         | The client's proposed budget, often informed by AI Price Prediction. |
| <b>Quote</b>     | amount                | float         | The worker's proposed price for the task.                            |
| <b>Review</b>    | rating                | float         | Raw numerical rating.                                                |
| <b>Review</b>    | sentiment_score       | sentimentEnum | AI-derived score (Positive, Negative, Neutral).                      |
| <b>AI Engine</b> | matchClientToWorker() | List<Worker>  | Primary method for AI-Powered Matching.                              |

## 4.4 Process Flow/Data Flow Representation

The **Process Flow Diagram (PFD)** maps the sequential user experience and the underlying data transitions, integrating the authentication, management, and real-time collaboration modules. This diagram functions as a high-level Data Flow Representation (DFD) for the system's operational logic.

### 4.4.1 Process Flow Diagram

The process begins with user authentication, leading to role-based access and task management, culminating in reporting and analytics.

This flowchart delineates the comprehensive operational workflow of the TaskIO platform, mapping the user journey from initial access to final reporting. It visualizes the logical progression

of tasks, decision points, and the interactions between different user roles (Clients, Workers, and Admins) within the system.

### **Detailed Workflow Phases:**

#### **1. Authentication Phase:**

- **Entry:** The process initiates with the "User Logs in with Credentials" action.
- **Validation Logic:** A decision diamond ("System Verifies Role") acts as the primary security gate.
  - **Invalid Path:** If verification fails, the system triggers a "Display Login Error" and loops the user back to "Restart Login".
  - **Valid Path:** Successful authentication grants access to the core application, transitioning the user to Task Management.

#### **2. Task Management & Hiring:**

- **Navigation:** Users land on the "Task Board / Dashboard" and proceed to "View Available Tasks" or their personal "My Tasks List".
- **Bidding Cycle:** The workflow includes a critical decision node where a "Worker Submits Quote." This step bridges the gap between discovery and engagement.
- **Assignment:** Once a quote is accepted, the "Client Hires Worker / System Assigns Task" process executes, officially linking a user to a project and updating their "My Tasks List".

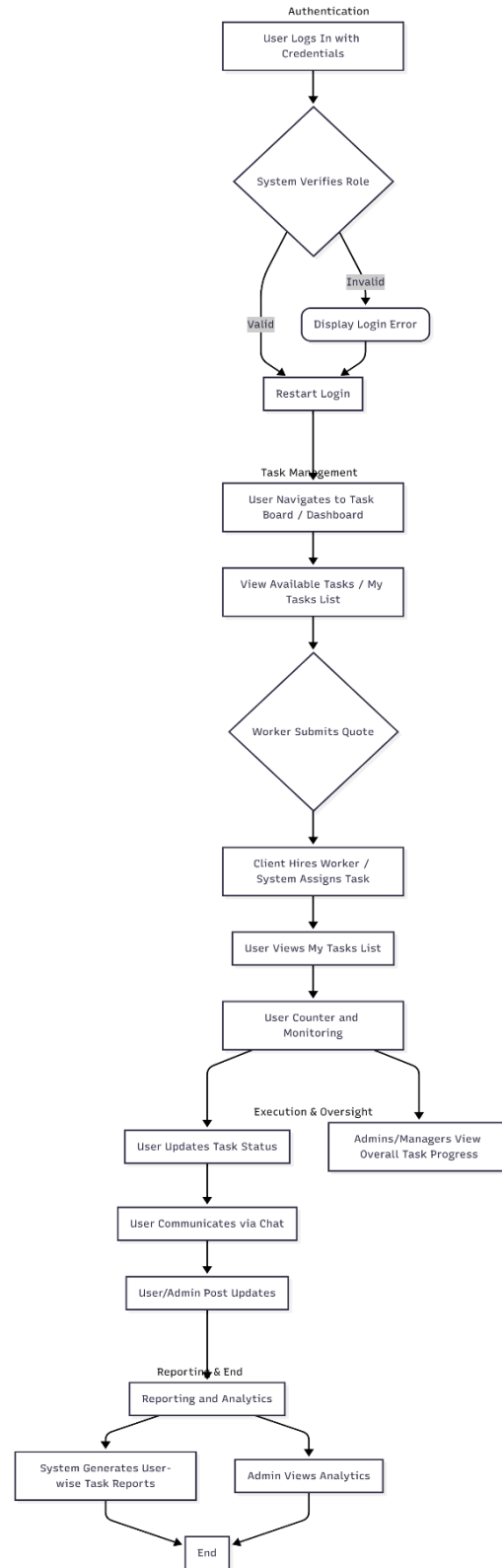
#### **3. Execution, Oversight & Communication:**

- **Monitoring:** The flow moves to "User Counter and Monitoring," where the path splits into parallel streams to demonstrate role-specific actions:
  - **Operational Stream (Left):** Users actively engaged in work perform actions like "User Updates Task Status," "User Communicates via Chat," and "Post Updates" to track progress.

- **Managerial Stream (Right):** Simultaneously, "Admins/Managers View Overall Task Progress," ensuring high-level oversight without interfering with daily operations.

#### **4. Reporting & Conclusion:**

- **Analytics:** Both streams converge at the "Reporting and Analytics" module.
- **Output:** The system generates tailored outputs: "User-wise Task Reports" for individual performance tracking and "Admin Views Analytics" for platform-wide health checks.
- **Termination:** The workflow concludes at the "End" state, signifying the completion of the user session or task lifecycle.



**Figure 4.5: Process Flow Diagram of Taskio**

The **Process Flow Diagram** is a visual representation of the core operational steps within the Taskio system, showing the movement of control and data through the platform's major functionalities, from user access to reporting.

The diagram is structured sequentially across three main phases:

1. **Authentication and Access:** The process begins with User Authentication, where credentials are used to verify the user's identity and determine their specific role (Admin/User). Access is either granted, leading the user to their appropriate dashboard, or denied with an error message.
2. **Task Management and Allocation:** Once access is granted, the user enters the core marketplace flow, beginning with Task Listing and Management. Users view available tasks, and the system handles the logic for assigning selected tasks to the worker after the Client performs the hiring action. This leads into User Monitoring, where status updates are tracked and managed (e.g., the worker updates the task status).
3. **Collaboration and Reporting:** This phase covers project execution and oversight. Users communicate via real-time chat, posting updates and collaborating to complete the task. The process culminates in the Reporting and Analytics section, where the system generates reports for users and provides administrative views on overall task progress and system health. The entire flow concludes after analytics are viewed, marking the end of the transaction lifecycle.

## 4.5 Design Models

In addition to the overall architecture, several UML models were developed to describe dynamic interactions and state management within the system.

### 4.5.1 Task State Diagram

The Task lifecycle is controlled by a Finite State Machine (FSM), ensuring that financial and project management integrity is maintained by allowing only defined transitions. This sequence diagram defines the secure workflow for task creation and moderation on the TaskIO platform. Unlike the standard user flow, this diagram emphasizes the governance role of the Administrator, ensuring that no content goes live without passing through a validation checkpoint. It maps the

interactions between two human actors (Client, Admin) and three system components (WebApp UI, Task Controller, Database).

### **Detailed Workflow Steps:**

#### **1. Secure Authentication (Client Side)**

- **Request:** The process initiates when the Client submits their login(credentials) through the WebApp UI.
- **Validation:** The Task Controller receives this request and triggers a Validate Login operation.
- **Verification:** The controller queries the Database to Check User records. Upon confirmation (OK), the system grants access and renders the Dashboard for the client.

#### **2. Task Submission & Data Persistence**

- **Submission:** The Client fills out the Post Task Form and submits it via the UI.
- **Processing:** The WebApp UI forwards this payload (Send Task Data) to the Task Controller.
- **Storage:** The Controller executes a Save Task(data) command to the Database, which stores the task with an initial status.
- **Confirmation:** The Database returns a Success signal, and the UI displays a Task Posted Confirmation to the Client.

#### **3. Asynchronous Admin Review (The Security Layer)**

- **Notification:** A critical backend process, Notify Admin (Asynchronous), is triggered immediately after the task is saved. This loop ensures the administration team is alerted without delaying the client's session.
- **Intervention:** The admin actor accesses the system to perform a Review Task operation, retrieving the data from the Database.

- **Decision:** The workflow concludes with the admin making an Approve? decision, which updates the task status in the database, effectively making it visible to workers or rejecting it for policy violations.

### Key Technical Features:

- **Role Separation:** The diagram clearly distinguishes between the client and the admin.
- **Asynchronous Processing:** The self-referencing loop on the Task Controller (Notify Admin) indicates that administrative alerts happen in the background, optimizing system performance.
- **Three-Tier Architecture:** The flow explicitly models the separation of concerns between the UI (Presentation), Task Controller (Logic), and Database (Persistence).

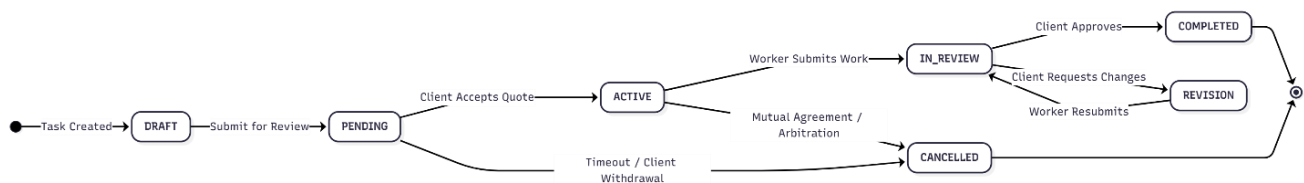


Figure 4.6: State Transition Diagram for the Task Entity

The **State Diagram for the Task Entity** visually defines the official lifecycle of a task (or 'order') on the Taskio platform, serving as a Finite State Machine to ensure transactional integrity and adherence to business rules.

The diagram illustrates the mandatory progression through key states and the specific triggers required for each transition:

- **DRAFT, PENDING:** A task starts with draft and transitions to pending once the Client submits the final details and budget, making it available for workers to submit quotes.
- **PENDING, ACTIVE:** This is the hiring moment. The status changes to activate when the Client formally accepts a worker's quote, and the project begins.
- **ACTIVE, IN\_REVIEW:** The status moves to review when the Worker submits the completed work, awaiting the Client's final assessment.

- **IN\_REVIEW, COMPLETED / REVISION:** This is the final decision point. The task is marked completed upon Client Approval, which triggers the final payment release from escrow. Alternatively, the Client can request revision, sending the task back to the worker.
- **Cancellation:** A task can be cancelled from either the pending or activate state, typically due to client withdrawal, expiration, or mutual agreement.

This model prevents unauthorized skips, ensuring that the critical step of payment release only occurs after the verified completed state is reached.

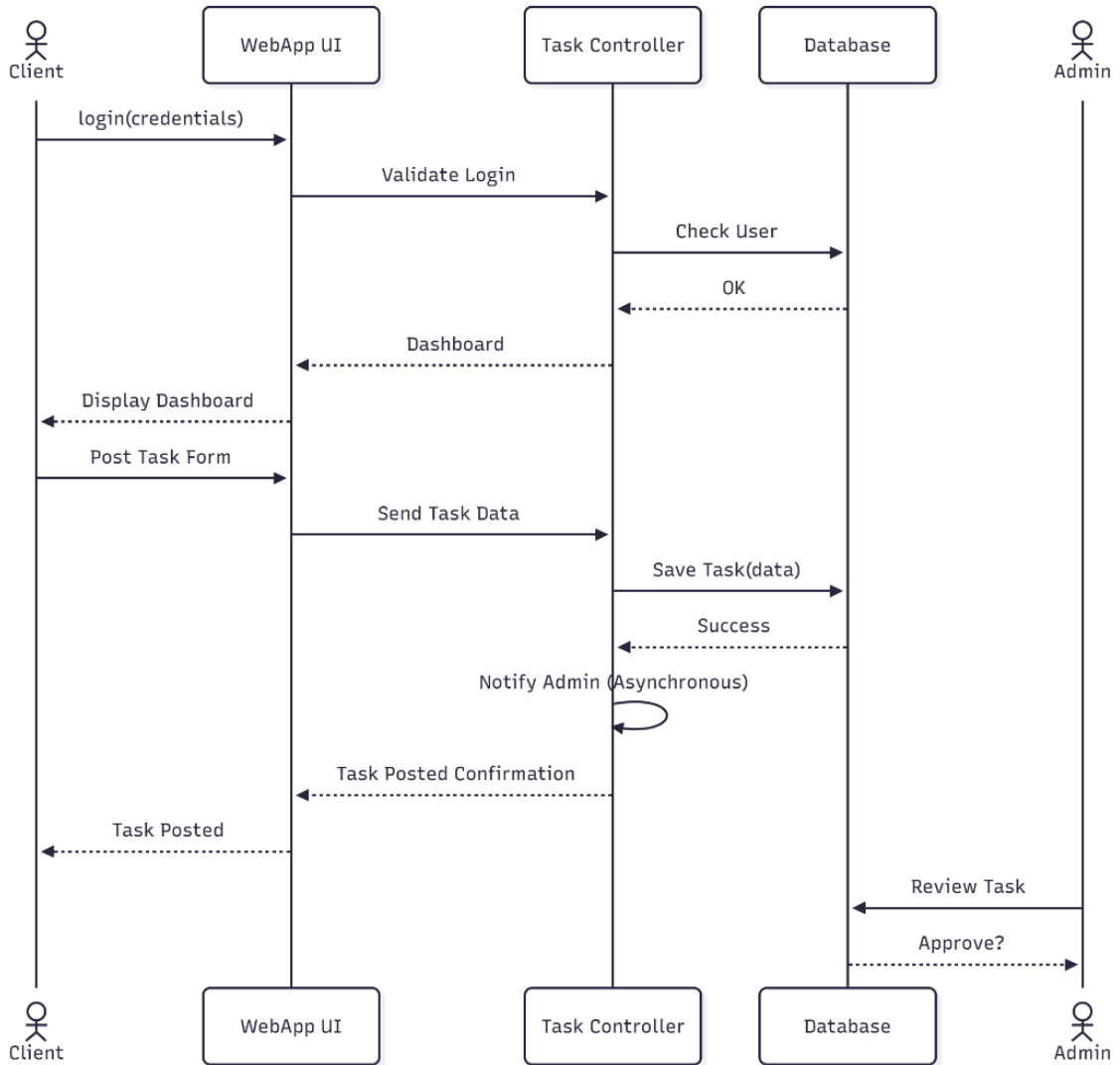
#### 4.5.2 Client/Task Posting Sequence Diagram

The **Sequence Diagram** illustrates the dynamic flow of messages and the chronological interaction between the major components during a client's core activities: user authentication and task posting.

The diagram maps the execution sequence across five key participants: the Client, the WebApp UI (presentation layer), the Task Controller (application/business logic layer), the Database (data layer), and the admin.

1. **Authentication Flow:** The sequence starts with the Client initiating login to the WebApp UI. The UI delegates the Validate Login request to the task controller, which checks the user's credentials against the Database. Upon successful validation (OK), the Task Controller sends the Dashboard response back to the client.
2. **Task Posting Flow:** After authentication, the Client requests the Post Task Form. The UI sends the filled Task Data back to the Task Controller. The Controller's core action is to persist with the data via the Save Task message to the Database.
3. **Oversight and Confirmation:** After the task is successfully saved, the Controller executes an asynchronous Notify Admin action. This highlights the platform's embedded oversight mechanism. Finally, the system confirms successful posting with a Task Posted message back to the Client.

The Sequence Diagram illustrates the dynamic interaction over time between the key objects and actors during the crucial task posting process.



**Figure 4.7: Sequence Diagram of Client/Task Posting**

### 4.5.3 Task Posting & AI Pricing Workflow Sequence Diagram

This sequence diagram illustrates the end-to-end interaction flow for a client user posting a new task on the TaskIO platform. It details the synchronous and asynchronous communication between the Client, the Web Interface, the Backend Services (Task Controller, AI Pricing Service), and the Database, culminating in the notification of relevant Workers.

#### 1. Authentication Phase

- **Initiation:** The workflow begins with the client requesting the login page.

- **Validation:** The user enters credentials (Email/Password), which are sent to the Task Controller. The controller validates these against the Database.
- **Authorization:** Upon successful verification (User Found & Verified), the system generates an Authentication Token and redirects the user to the Dashboard, establishing a secure session for subsequent actions.

## 2. Task Posting & AI Integration

- **Input:** The Client navigates to the "Post a Task" form and enters the Title, Description, and Category.
- **AI Trigger:** A specific event (Blur Event) on the description field triggers the intelligent pricing logic before the form is even submitted.
- **AI Processing:** The Web Interface asynchronously sends the task description and category to the AI Pricing Service.
- **Regression Analysis:** The AI service executes a Calculate Market Rate (Regression Model) internal process to analyze the task complexity against market data.
- **Feedback:** The service returns a Suggested Price Range, which is dynamically displayed to the user to guide their budget setting.

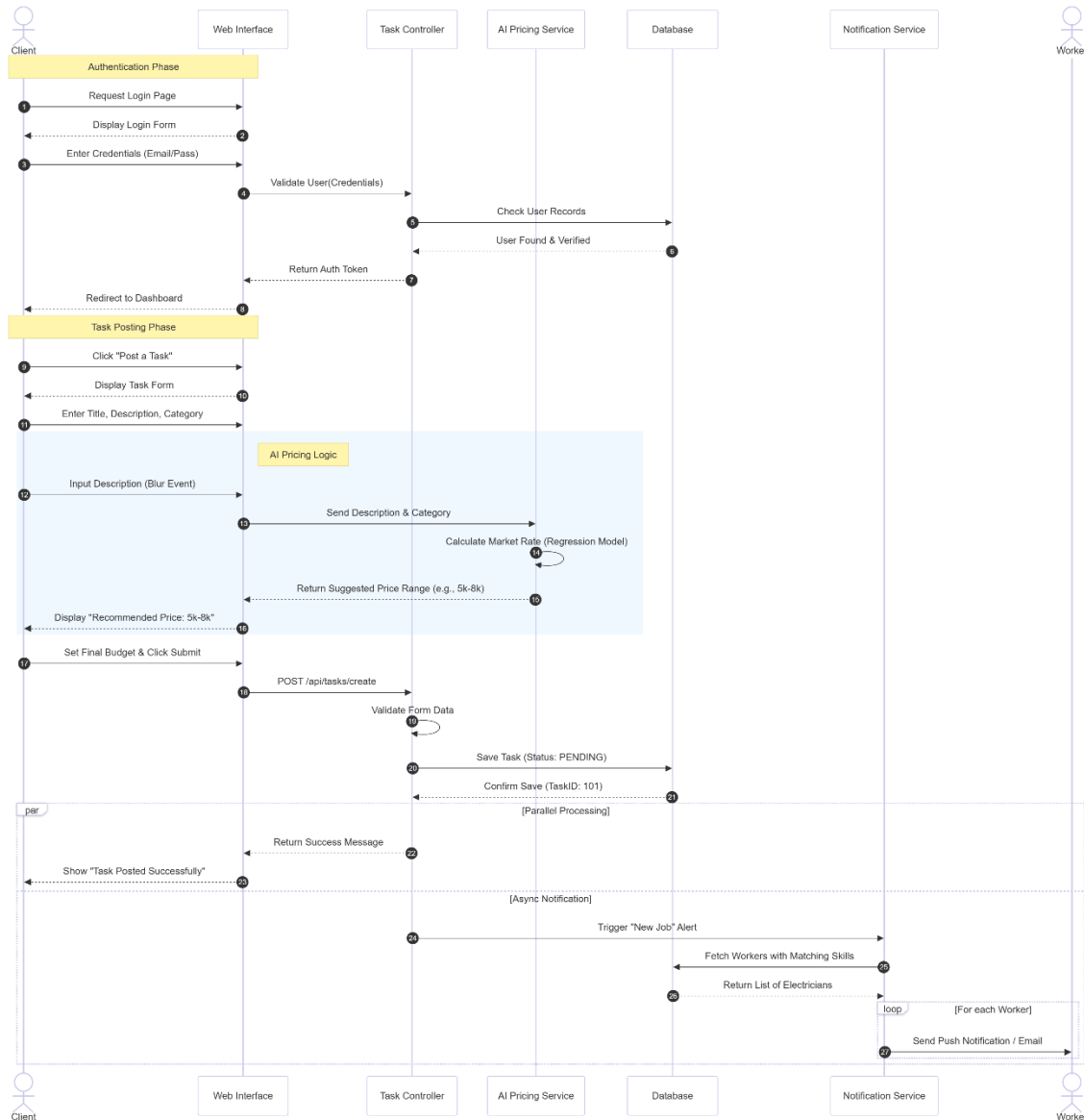
## 3. Submission & Data Persistence

- **Finalization:** The Client sets the final budget (possibly influenced by the AI suggestion) and submits the form.
- **Persistence:** The Task Controller validates the form data and commits the new task to the **Database** with a status of pending.

## 4. Asynchronous Notification System

- **Parallel Execution:** To ensure a responsive user experience, the system splits into a parallel process (using a par block).
  - **User Feedback:** The system *immediately* returns a "Success Message" to the Client, allowing them to continue using the app without waiting.

- **Background Notification:** Simultaneously, the notification Service is triggered. It queries the database to Fetch Workers with Matching Skills.
- **Loop:** The system iterates through the list of identified workers and sends a Push Notification / Email to each, alerting them of the new opportunity.



**Figure 4.8: Sequence diagram of Task Posting & AI Pricing Workflow**

# **Chapter 5**

## **Implementation**

## 5 IMPLEMENTATIONS

### 5.1 Algorithm

The implementation of the system's intelligence is centralized within the AI Engine utility class, which encapsulates all complex mathematical and machine learning logic. These critical features are executed asynchronously using Celery workers and a message broker to ensure high performance and prevent degradation of the platform's real-time responsiveness.

#### 5.1.1 AI Task-to-Freelancer Matching Algorithm

The AI Matching Algorithm is a hybrid recommendation system designed to accurately pair a newly posted client task with the most qualified and reputable workers available in the platform's 'stock' of talent. This approach moves beyond simple keyword matching by synthesizing multiple data points to generate a comprehensive compatibility score.

The process begins by calculating the Semantic Relevance between the task's detailed description and a worker's profile information using pre-trained language models, effectively determining how well the worker's narrative aligns with the job's context. Concurrently, the algorithm assesses the Skill and Performance Alignment, which measures the direct overlap of required versus stated skills, alongside a heavily weighted analysis of the worker's historical behavior. This historical data includes quantifiable metrics like job completion rates and a composite reputation score derived from past customer reviews, which is augmented by sentiment analysis. The final match score is a blended weighted average of these components, ensuring that high skill is prioritized, but only when coupled with a proven record of reliability and positive feedback. Workers receiving the highest match scores are then prioritized for immediate notification regarding the new task, streamlining the client's hiring process.

#### 5.1.2 Dynamic Pricing Recommendation Algorithm

To address the lack of pricing transparency in the market, an AI-based dynamic pricing model was implemented to recommend an optimal budget range to the client when posting a new job. This feature relies on a supervised regression model trained in historical task data.

When a client inputs a new task, the model processes several attributes of the proposed task in real-time, such as its stated category, geographical location, estimated complexity (derived from description length and keywords), and the current market rates for similar completed projects. The model executes its prediction and returns a single optimal price estimate. To provide flexibility to the client, a narrow confidence band, or range, is calculated around this predicted price, presenting both a suggested minimum and maximum acceptable budget. This process, encapsulated in the `calculateDynamicPricing()` method, guides the client toward setting a fair and competitive budget, ultimately reducing negotiation friction and encouraging skilled workers to submit competitive quotes.

### **5.1.3 Sentiment Analysis of Feedback**

The platform integrates sentiment analysis to provide a deeper, qualitative layer to the worker reputation system, moving beyond basic star ratings.

This algorithm takes the raw text content of a client's review and processes it using specialized Natural Language Processing (NLP) techniques. The system analyzes the text's emotional tone to determine the underlying sentiment (e.g., Positive, Negative, or Neutral) and assigns a quantifiable sentiment score. This score is then saved within the associated Review entity, providing an objective metric of the quality-of-service delivery. This feature, accessible via the `performSentimentAnalysis()` method, ensures that the worker's reputation profile is a robust measure that includes the nuance of client feedback, directly feeding into the accuracy of the AI Matching system.

## **5.2 External APIs**

Seamless communication and secure financial transactions are enabled by integrating with dedicated external service providers, ensuring the platform remains localized, secure, and compliant.

### **5.2.1 Payment Gateway Integration**

To facilitate reliable and secure financial transactions, the system integrates with both international and essential local payment processors. The payment module supports Stripe API for international card transactions, while also incorporating local payment methods critical for the Pakistani market,

such as JazzCash API and EasyPaisa API. This integration allows the platform to manage the escrow system functionality, where funds are securely held upon task assignment and released to the worker only upon the client's final approval of the completed work, thus providing financial trust to both parties.

### 5.2.2 Communication and AI Services

For robust asynchronous communication and real-time features, the following integrations were established:

- **Email Notification Service:** A transactional email service API is integrated to handle high-volume communications, including account verification during signup, password resets, and critical status changes throughout the task lifecycle.
- **Voice Assistance:** The system has API hooks to integrate with advanced language models, providing AI-powered translation services to automatically translate messages between clients and workers.
- **Identity Verification:** The system design includes components for image and text processing APIs, intended to support the future implementation of automated CNIC image validation using OCR/FaceNet components. This is a critical step for enhancing trust and security within the marketplace.

**Table 5.7: Details of APIs used in the project**

| Name of API       | Description of API                                               | Purpose of API                                                                                          | List down the function/class name in which it is used                                  |
|-------------------|------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| <b>Stripe API</b> | This international platform is used for processing card payments | It handles secure global payments and manages the financial transactions within the task escrow system. | Payment and Billing Service; Internal methods within apps/payments/payment_gateway.py. |

|                                        |                                                                                                                 |                                                                                                                                         |                                                                                                                                           |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| <b>JazzCash API / EasyPaisha API</b>   | These are the local payment gateways essential for digital wallet transactions in Pakistan.                     | The integration ensures platform localization and provides necessary financial accessibility for the local target market.               | Payment and Billing Service; Internal methods within apps/payments/payment_gateway.py.                                                    |
| <b>OpenAI / Custom ML Hosting API</b>  | These external service providers offer scalable infrastructure for hosting and running Machine Learning models. | They enable complex AI features, including matching, dynamic pricing, sentiment analysis, and translation.                              | AIEngine.matchClientToWorker(), AIEngine.calculateDynamicPricing(), AIEngine.performSentimentAnalysis(), and AIEngine.translateMessage(). |
| <b>Email Service API</b>               | This is a dedicated platform for sending high-volume, automated transactional emails.                           | It ensures secure communication, specifically sending verification links during registration and notifications for task status changes. | Internal service methods within apps/users/email_service.py.                                                                              |
| <b>CNIC Image Verification Service</b> | This third-party service utilizes OCR and image processing technology.                                          | It helps to improve user trust and verify worker credentials by processing uploaded images of identity documents.                       | Verification Service; Internal methods within apps/users/identity_verification.py.                                                        |

## 5.3 User Interface

The User Interface (UI) is the presentation layer designed for maximum usability and seamless user experience, implemented using responsive web design to ensure compatibility across all devices.

### 5.3.1 Authentication Views (Login, Signup)

The Login and Signup views are streamlined to prioritize security and clarity. The Registration Interface clearly prompts the user to select their role (Client or Worker) before proceeding, which is foundational to the platform's role-based permissions. Secure password policies are explicitly displayed during the signup process to ensure compliance with security requirements. The Login Interface is minimalist, focusing on credentials and providing simple navigation for password recovery.

### 5.3.2 Task Posting and Ordering Interface

The Task Posting Interface (/tasks/post) is designed to walk the client through the creation of a detailed task 'order.' This view is critical as it presents the output of the Dynamic Pricing Algorithm. After the client enters the required descriptive information, the system dynamically displays the AI-generated budget recommendation. This feature guides the user toward an optimal price range, which is a key mechanism for ensuring fair market practices and client satisfaction.

### 5.3.3 Dashboard and Stock Management View

Upon successful login, users are redirected to their personalized dashboard, which serves as the central hub for all project-related activities.

- **Client Dashboard:** Primarily displays the status of all posted tasks ('orders'), received quotes, and payment/escrow management views.
- **Worker Dashboard (Stock View):** Functions as the primary interface for workers to discover new work. It displays a curated list of available tasks, effectively acting as the inventory or 'stock' of opportunities. Crucially, this list is prioritized and ranked according to the AI Matching Algorithm results, ensuring the worker sees the most relevant jobs first, maximizing efficiency and minimizing time wasted searching irrelevant listings.

### 5.3.4 Screenshot of Login Screen

**TaskIO Login Interface** The screenshot demonstrates the fully implemented Login Interface for the TaskIO web platform, currently running in a local development environment. The page utilizes a modern split-screen layout designed to balance visual appeal with functionality. The left panel features a branded gradient background that reinforces the platform's value propositions, such as "Trusted Network" and "Secure Payment," to instill user confidence during access. The right panel contains a clean, centered authentication card where users input their credentials, complete with "Remember me" functionality and password recovery options. The top navigation bar provides immediate access to core modules like "Find Work" and "Find Talent." Additionally, the interface highlights the successful integration of the support system, visible as a floating AI chatbot widget in the bottom-left corner, ensuring real-time assistance is available from the moment a user lands on the application.

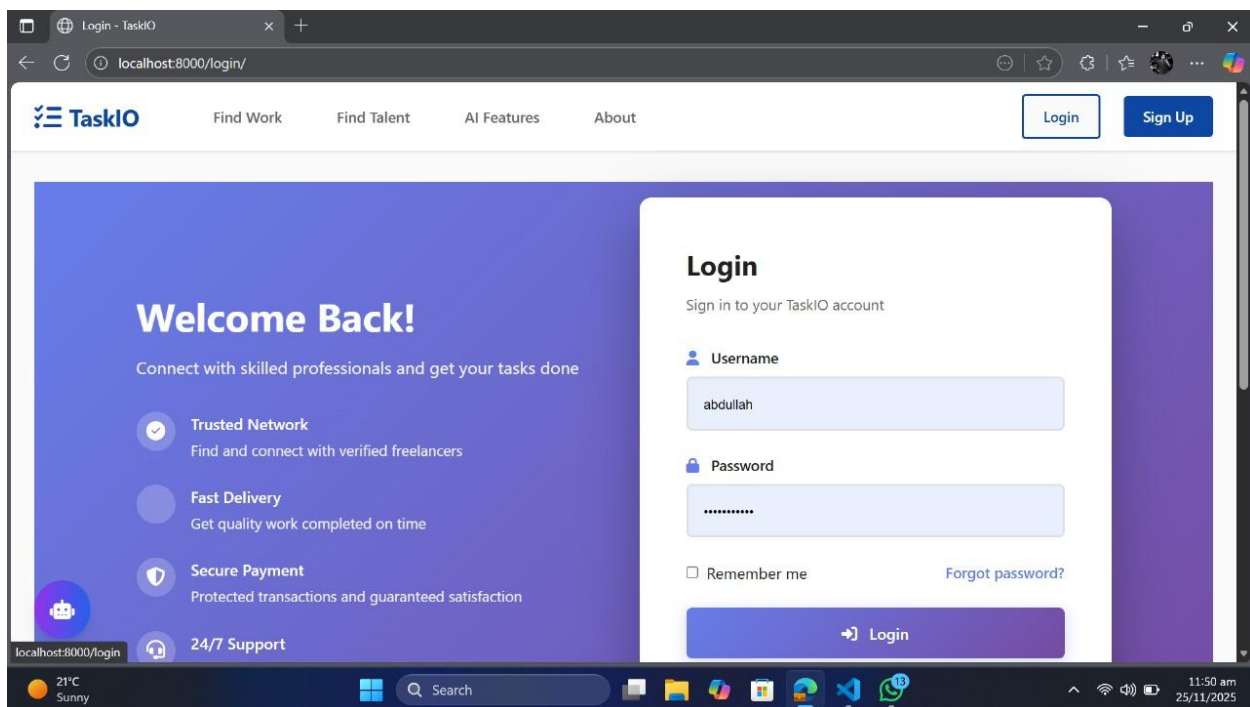


Figure 5.9: Screenshot of Login Screen

### 5.3.5 Screenshot of Signup Screen

TaskIO User Registration Interface This figure illustrates the "Sign Up" module of the TaskIO platform, designed to streamline the onboarding process for new users. The interface maintains visual consistency with the application's branding by utilizing the primary purple color scheme and a clean, centered card layout. The registration form captures essential user data, featuring fields for Username, Email address, Password, and Password Confirmation to ensure account security. Notably, the UI includes instructional helper text below the Username field to guide users on character limits and formatting requirements, minimizing input errors. The top navigation bar remains persistent, allowing easy switching between the "Login" and "Sign Up" views. Additionally, the AI chatbot widget in the bottom-left corner is available throughout the registration process, providing immediate assistance to users who may encounter issues while creating their accounts.

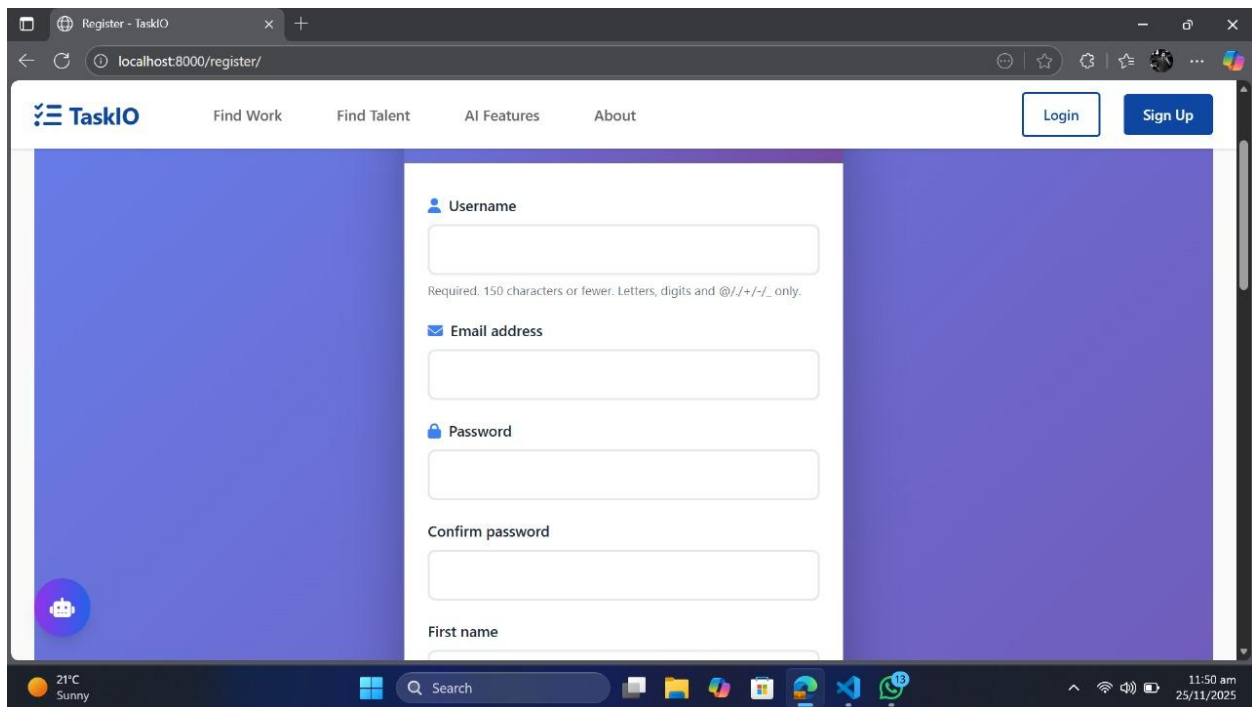


Figure 5.10: Screenshot of Signup Screen

### 5.3.6 Screenshot of User Home Screen

Taskio User Dashboard This figure displays the personalized User Dashboard of the Taskio application, serving as the central hub for managing user activities. The interface features a prominent "Welcome Back" hero section with a personalized greeting ("Abdullah Rafiq"), creating an engaging user experience. Key performance metrics are visualized through a clean card-based layout, displaying real-time data such as "Active Tasks," "Pending Applications," "Total Earnings" (displayed in PKR), and "Average Rating." The top navigation bar has been updated to include authenticated user controls, such as notification alerts, messages, and a profile dropdown menu. Below the main banner, the summary statistics are repeated with distinct iconography and color coding (e.g., green for tasks, blue for earnings) to improve visual scalability. Consistent with the platform's design, the AI chatbot widget remains accessible in the bottom-left corner to assist users with dashboard navigation.

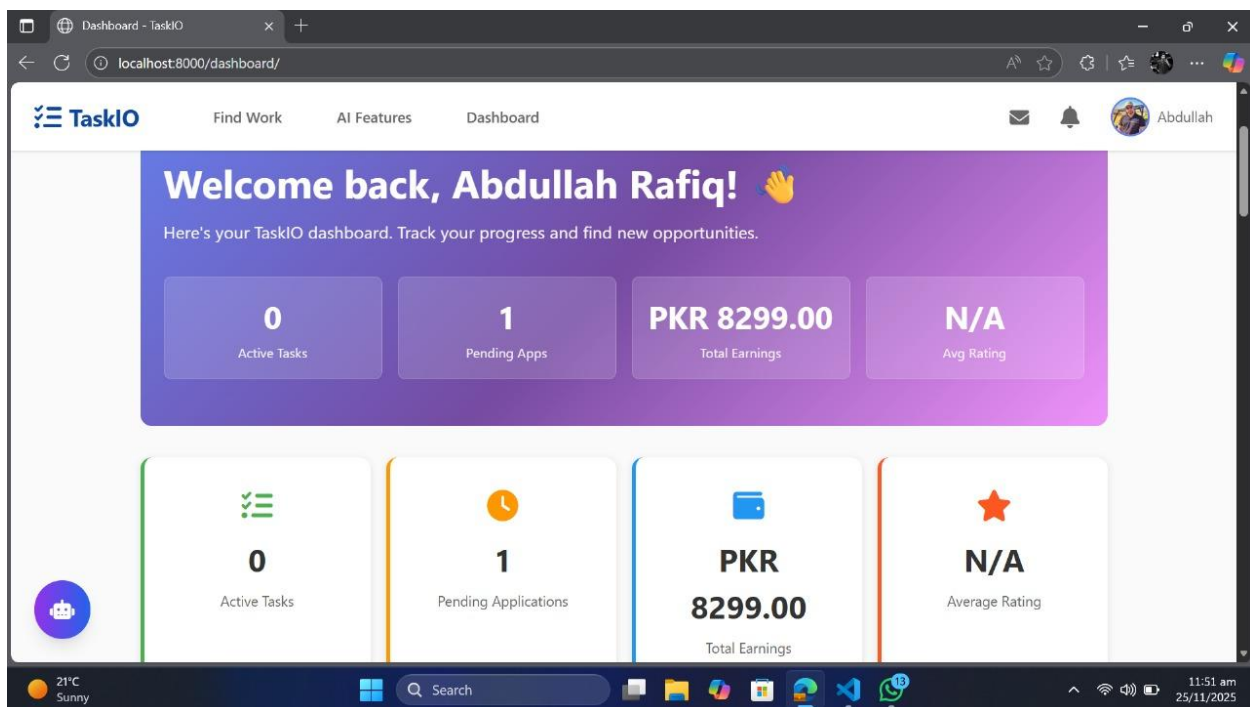


Figure 5.11: Screenshot of User Home Screen

### 5.3.7 Screenshot of User Analytics & Performance Visualization

TaskIO Analytics & Performance Visualization This figure highlights the data visualization module within the TaskIO User Dashboard, designed to provide users with actionable insights into their freelancing performance. The interface features two primary graphical components arranged in a clean, card-based layout. The "Earnings Trend" chart on the left utilizes a line graph with area shading to track financial progress over time, clearly visualizing income spikes across a six-week period. To the right, the "Task Distribution" donut chart offers a categorical breakdown of task statuses, using distinct color coding for "Accepted" (green), "Pending" (blue), and "Rejected" (orange) applications. This visual hierarchy allows users to instantly assess their success rates and workflow status. Consistent with the application's design, the global navigation bar and the floating AI support widget remain accessible, ensuring a seamless user experience while reviewing performance metrics.

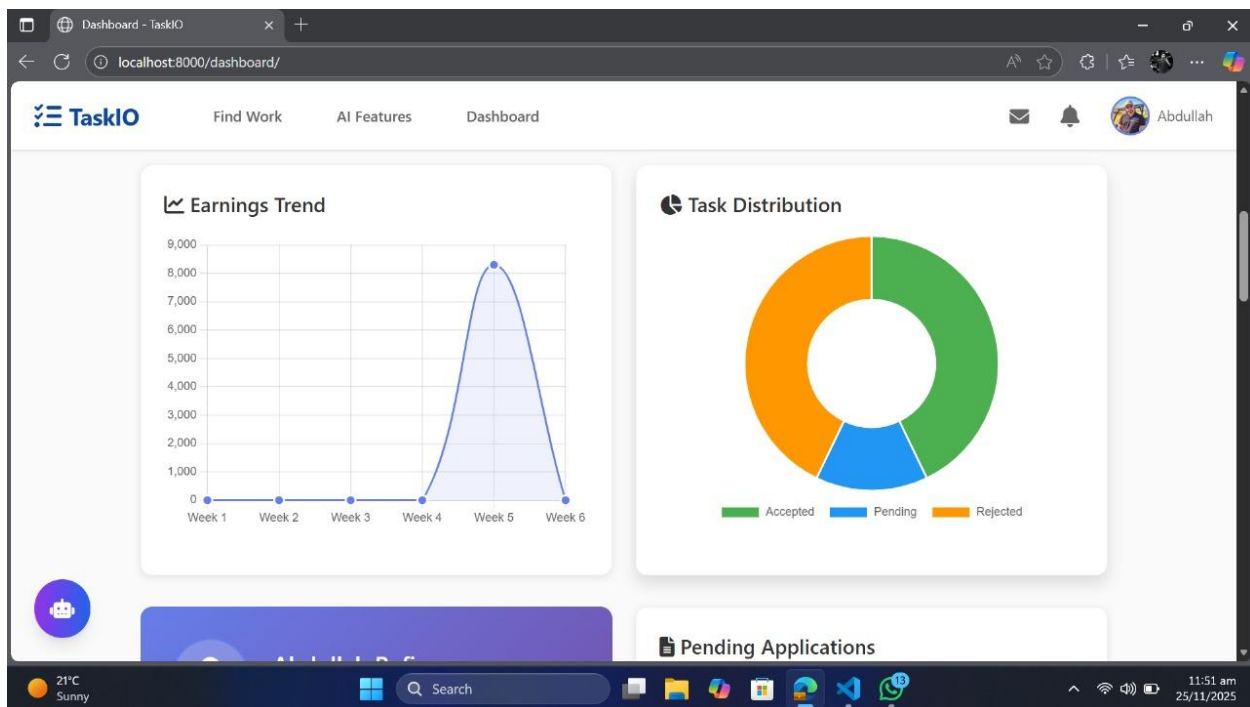


Figure 5.12: Screenshot of User Activity

### 5.3.8 Screenshot of User Profile interface

**TaskIO User Profile Interface** This figure illustrates the User Profile module of the TaskIO platform, designed to present a comprehensive view of the user's identity and account status. The layout utilizes a modular card-based design to organize personal information effectively. The primary section showcases the user's identity, featuring a profile avatar, display name ("Abdullah Rafiq"), and location, alongside status indicators such as "Freelancer" and account verification tags. An "Edit Profile" button is prominently placed to facilitate immediate updates to user data. To the right, dedicated widgets display contact information—including email and phone number—and account history, specifically the "Member Since" date. The interface maintains visual consistency with the rest of the application through uniform styling, a persistent navigation bar, and the always-accessible AI chatbot widget in the bottom-left corner.

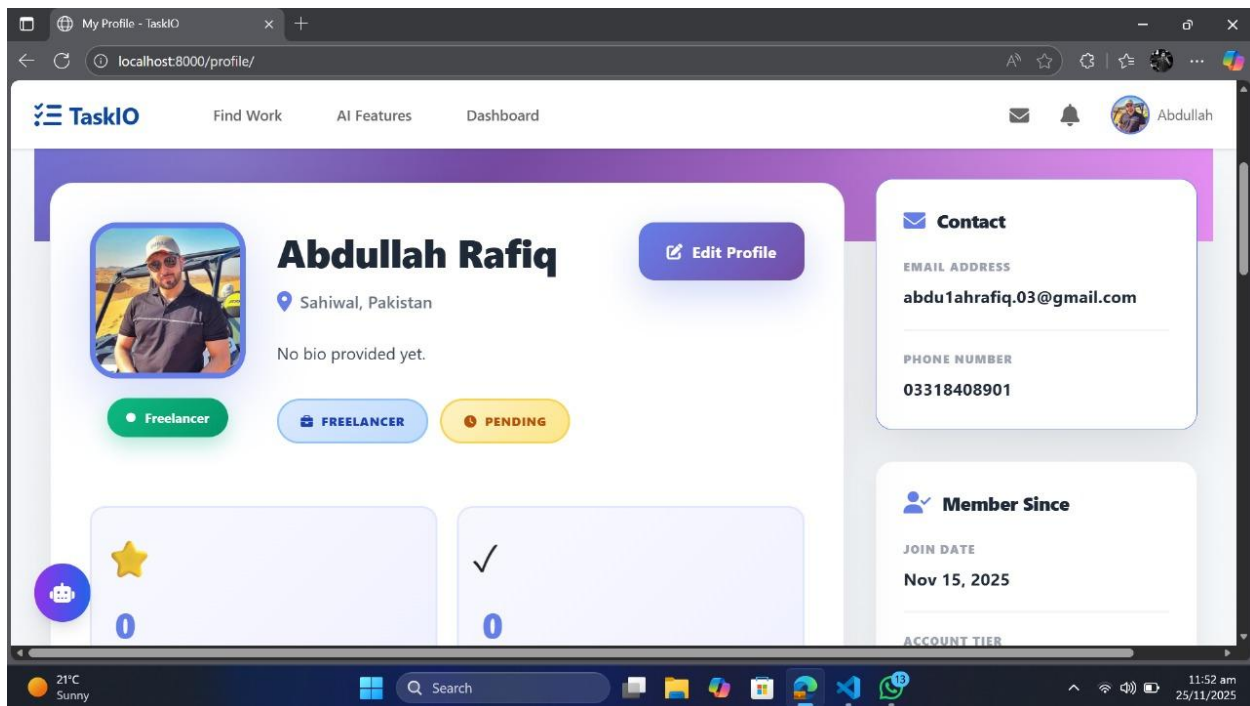
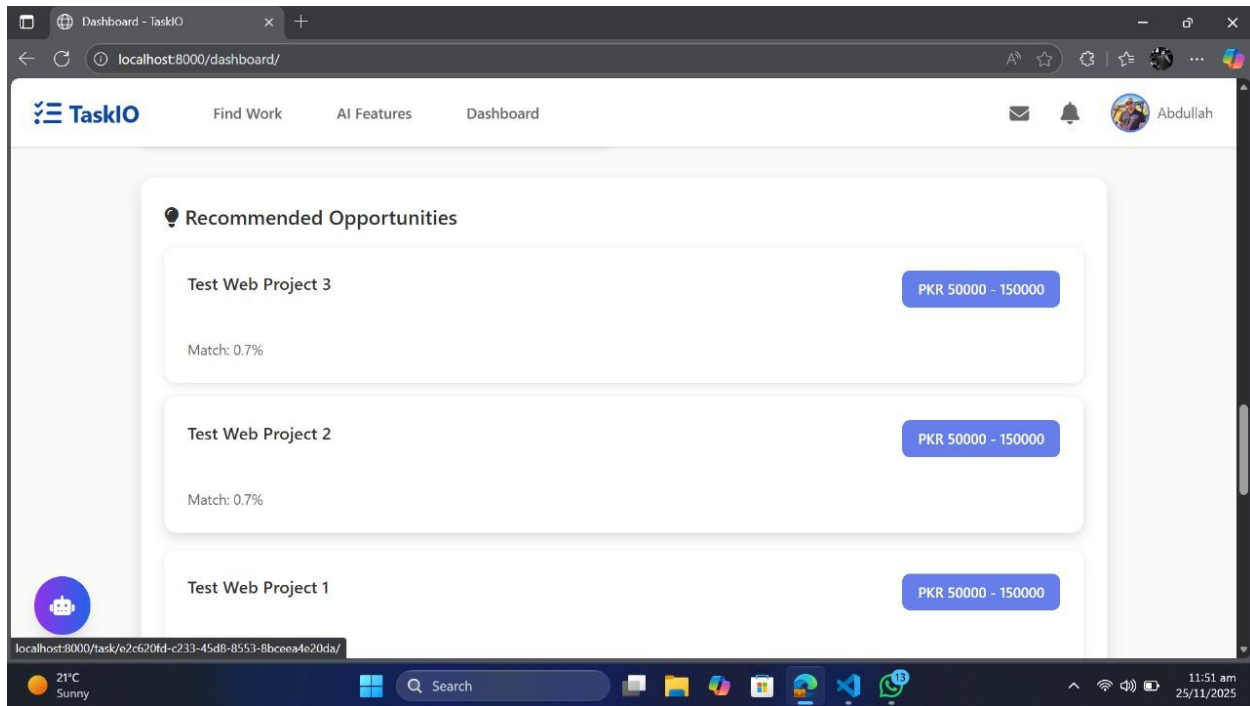


Figure 5.13: Screenshot of User Profile interface

### 5.3.9 Screenshot of Recommendation System interface

**TaskIO Recommended Opportunities Interface** This figure depicts the "Recommended Opportunities" module, a core component of the TaskIO dashboard designed to facilitate intelligent job matching. The interface displays a dynamic list of project cards, such as "Test Web Project 3," which are curated based on the user's profile and skills. Each card features a clean layout presenting key decision-making data: the project title, a prominent budget range button (displayed in PKR), and a specific "Match" score (e.g., 0.7%). This metric visualizes the system's backend logic, helping users prioritize applications where they have the highest probability of success. The standardized design ensures consistency with the platform, utilizing the primary blue color scheme for call-to-action elements. As with other screens, the AI chatbot widget is anchored in the corner, ready to assist users in navigating these opportunities or understanding their match scores.



**Figure 5.14: Screenshot of Recommendation System interface**

### 5.3.10 Screenshot of Proposal Submission interface

TaskIO Proposal Submission & AI Pricing Interface This figure displays the "Apply for Project" interface, a critical module where freelancers submit bids for specific tasks. The screen is divided into a main input section and an informational sidebar. The primary form captures key proposal details, including "Proposed Price" and "Proposed Duration," with clear indicators of the client's budget range. A distinguishing feature of this interface is the integrated "AI Price Prediction" widget. This blue panel utilizes market analysis algorithms to suggest a competitive rate (e.g., PKR 4633.58), aiding freelancers in optimizing their bids for higher success rates. The right-hand sidebar provides essential context, including a client profile summary and a "After Applying" guide that outlines the subsequent workflow steps. As consistent with other modules, the AI chatbot widget is positioned in the corner to offer real-time support during the proposal drafting process.

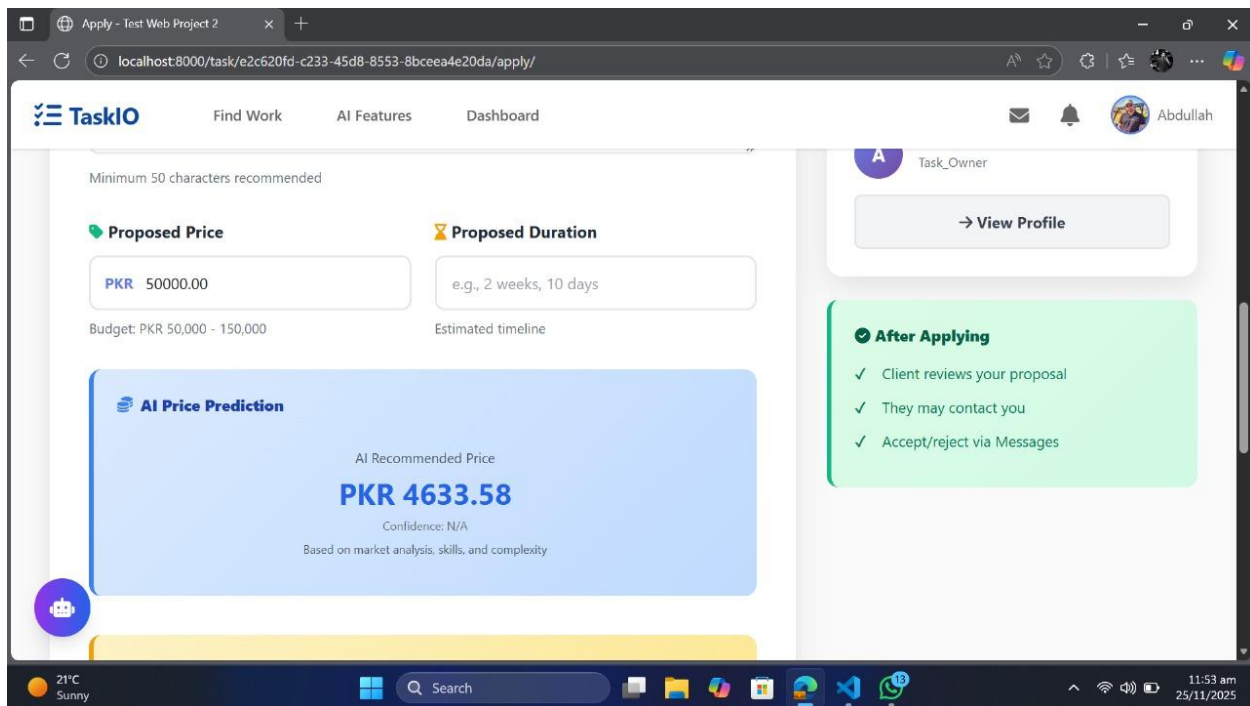


Figure 5.15: Screenshot of Proposal Submission interface

### 5.3.11 Screenshot of Messaging/Chat System interface

**TaskIO Real-Time Messaging Interface** This figure illustrates the integrated Messaging Module, designed to facilitate seamless communication between freelancers and clients directly within the platform. The interface utilizes a familiar two-column layout: the left sidebar organizes active conversations with a search bar for quick retrieval and a "New Message" trigger, while the right panel displays the active chat thread. The conversation view visually distinguishes between incoming (white) and outgoing (branded gradient) messages for clarity. Key features include real-time status indicators (e.g., "Online"), timestamping for audit-trails, and a rich media input field that supports text, file attachments, emojis, and voice notes. This module ensures users can negotiate terms and collaborate securely without needing external communication tools. The persistent navigation bar and AI support widget are retained to maintain a cohesive user experience.

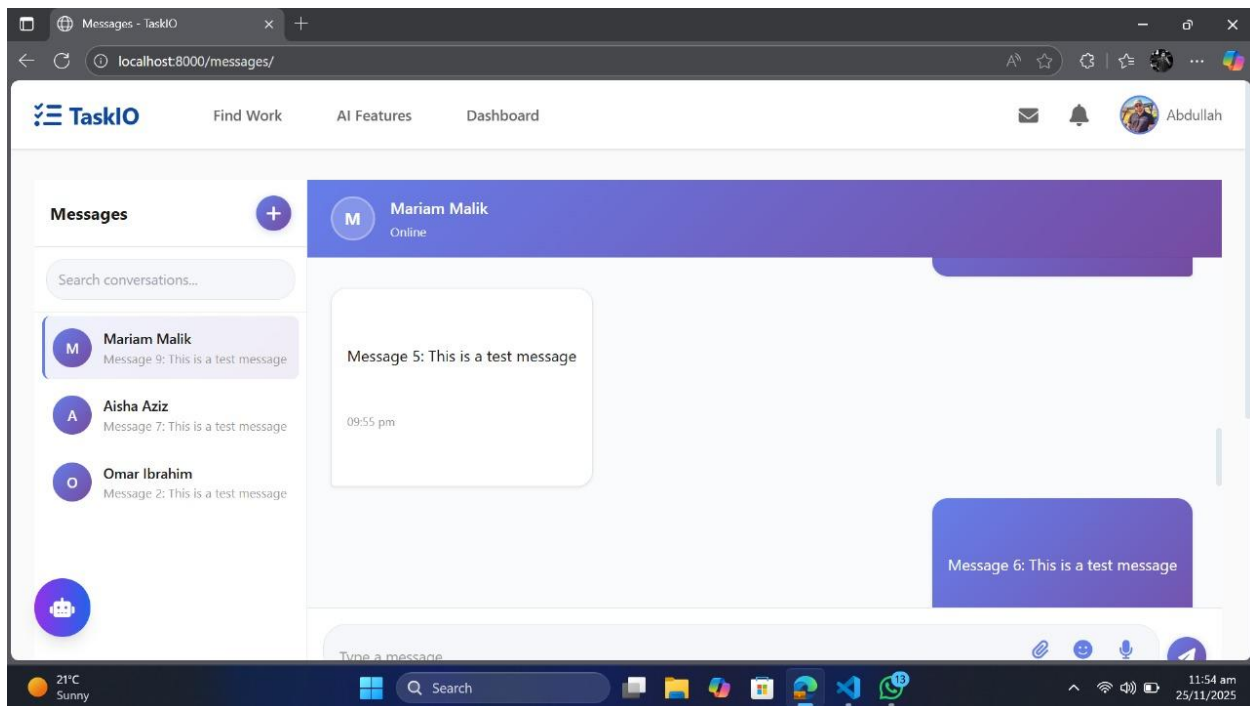


Figure 5.16: Screenshot of Messaging/Chat System interface

# **Chapter 6**

## **Testing and evaluation**

## 6 TESTING AND EVALUATION

### 6.1 Manual Testing

Manual testing was systematically performed across the entire system, focusing on validating core business logic, user experience, and the correct behavior of the integrated AI features that drive the marketplace's primary value.

#### 6.1.1 System Testing

System testing was performed on the fully integrated application to confirm that all modules interact correctly, and the entire system operates according to the overall design specifications and security standards

**Table 6.8: Manual Testing**

| No. | Test Objective                         | Test Script                                                                                                                                                                             | Expected Result                                                                                                                                                 | Result |
|-----|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| 1.  | Verify End-to-End Task Lifecycle.      | Client posts task (Order)<br>\$\\rightarrow\$ Worker applies (Quote)<br>\$\\rightarrow\$ Client hires<br>\$\\rightarrow\$ Worker submits<br>\$\\rightarrow\$ Client approves (Payment). | Task status transitions correctly (PENDING<br>\$\\rightarrow\$ ACTIVE<br>\$\\rightarrow\$ IN_REVIEW<br>\$\\rightarrow\$ COMPLETED).<br>Payment log is recorded. | Pass   |
| 2.  | Verify Cross-Browser/Device Usability. | Access all main pages (Dashboard, Task Listing) on Chrome/Firefox and resize to mobile screen widths.                                                                                   | UI remains consistent, navigable, and free of broken elements.                                                                                                  | Pass   |
| 3.  | Validate AI Integration Decoupling.    | While an AI Matching task is running (high-load scenario), perform an independent Task Posting.                                                                                         | Task Posting response time of the task is immediate (non-blocking) due to Celery offloading.                                                                    | Pass   |

## 6.1.2 Unit Testing

Unit testing focused on isolating and validating the smallest testable components, such as individual functions, data model methods, and security logic.

### Unit Testing 1: User Authentication and Role-Based Login

Testing Objective: To ensure the login process is secure, validates credentials, and correctly assigns the user to the appropriate role-based dashboard.

**Table 6.9: User Authentication**

| No. | Test Case/Test Script                                          | Attribute Value                                                    | Expected Result                                                                                             | Result |
|-----|----------------------------------------------------------------|--------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|--------|
| 1.  | Verify user login with correct input data (Client role)        | Username:<br>client_user@taskio.com<br>Password:<br>SecurePass123  | Successfully logs into the main Client Dashboard page.                                                      | Pass   |
| 2.  | Attempt login with a correct username but an invalid password. | Username:<br>worker_user@taskio.com<br>Password:<br>WrongPassword! | Login fails, and a generic error message ("Invalid Credentials") is displayed.                              | Pass   |
| 3.  | Verify role segregation upon login (Worker role).              | Username:<br>worker_user@taskio.com<br>Password:<br>WorkerPass123  | Successfully logs into the Worker Dashboard, which displays the prioritized Stock View (Recommended Tasks). | Pass   |

### Unit Testing 2: AI Dynamic Pricing Model Integrity

Testing Objective: To ensure the core AI feature for Dynamic Pricing provides accurate and logical price recommendations based on task complexity.

**Table 6.10: AI Dynamic Pricing Model Integrity**

| No. | Test Case/Test Script                             | Attribute Value                                                                                                                        | Expected Result                                                                                          | Result |
|-----|---------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|--------|
| 1.  | Test price prediction for a low-complexity task.  | The AI model returns to a low-end price range reflective of minimal effort.                                                            | The AI model returns to a low-end price range reflective of minimal effort.                              | Pass   |
| 2.  | Test price prediction for a high-complexity task. | Title: "Develop a full Django REST API microservice with DB migration." Category: Software Engineering. Description Length: 150 words. | The AI model returns to a high-end price range, commensurate with market value.                          | Pass   |
| 3.  | Validate model's adherence to price floor.        | Input that would typically result in a near-zero or negative price (e.g., invalid keywords).                                           | The system caps the minimum suggested price on the defined platform floor to prevent market devaluation. | Pass   |

### 6.1.3 Functional Testing

Functional testing validated that specific core functionalities and AI features conform precisely to the documented functional requirements, particularly focusing on the transactional flow.

**Table 6.11: Functional Testing**

| No. | Test Case/Test Script                 | Attribute and value                                                   | Expected Result                                                             | Result |
|-----|---------------------------------------|-----------------------------------------------------------------------|-----------------------------------------------------------------------------|--------|
| 1.  | Verify AI Matching of available Stock | Client posts Task: "Looking for an Electrician to fix office wiring." | The top 5 workers in the notification list are designated 'Electrician' and | Pass   |

|    |                                                                           |                                                                                                 |                                                                                                                                       |      |
|----|---------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|------|
|    | (Talent Pool) to a new Order.                                             |                                                                                                 | have a high completion rate/rating.                                                                                                   |      |
| 2. | Test Sentiment Analysis integrates into a negative review. (FR-09, TC-10) | Client submits a review: "Job completed late and communication was terrible. Do not recommend." | The system calculates a Negative sentiment score. This score is saved in the Review entity and lowers the worker's reputation weight. | Pass |
| 3. | Validate Real-time Messaging system.                                      | Two users (Client and Worker) open the chat interface simultaneously and exchange messages.     | Messages appear in both user interfaces instantaneously via the WebSocket connection.                                                 | Pass |

### 6.1.4 Integration Testing

Integration testing verified the seamless flow of data and control between the major system modules.

**Table 6.12: Integration Testing**

| No. | Test Case/Test Script                            | Attribute and value              | Expected Result                                                                                                                                | Result |
|-----|--------------------------------------------------|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| 1.  | End-to-End Task Posting and Worker Notification. | Client posts a Task.             | AI Matching Microservice (Celery) runs successfully and triggers the Email Service API. The worker receives an application notification email. |        |
| 2.  | End-to-End Task Posting and Worker Notification. | Client accepts a Worker's quote. | Task status changes to ACTIVE. The Payment and Billing Service is successfully called, initiating the fund transfer                            |        |

|    |                                        |                                                                      |                                                                                                                                              |  |
|----|----------------------------------------|----------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|--|
|    |                                        |                                                                      | to the Escrow account using the specified payment API.                                                                                       |  |
| 3. | Task Completion and Payment Release.   | Task Status is changed from IN_REVIEW to completed.                  | The Payment Service releases the funds from Escrow to the Worker's account via the integrated API. Both users receive an email notification. |  |
| 4. | Admin Action and Security Enforcement. | Admin flags and bans a fraudulent worker account (banUser() method). | The worker is immediately logged out and attempts to login or access protected routes are denied, enforcing role-based security.             |  |

## 6.2 Automated Testing

Automated testing is crucial for ensuring the platform's long-term maintainability and scalability, forming the foundation of the Continuous Integration/Continuous Deployment (CI/CD) pipeline. These tests are executed automatically upon code committees to guarantee that new changes do not introduce regressions.

**Table 6.13: Automated Testing**

| Tool Name                   | Tool Description                                                                                | Applied on                                                                                                                        | Results                                                                   |
|-----------------------------|-------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| Pytest / Django Test Client | Python's standardized testing framework and Django's built-in tool for simulating API requests. | Unit Tests for all core Model methods. Integration Tests for API endpoint responses. Functional Test for AI Matching Score Logic. | Verifies transactional integrity and security adherence across API layer. |

|                                           |                                                                                                         |                                                                                                            |                                                                                                       |
|-------------------------------------------|---------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Celery<br>Eager<br>Mode Tests             | A configuration used to execute asynchronous Celery tasks synchronously during the testing environment. | AI Feature Execution.<br>Performance validation of task execution. Email Notifications during user signup. | Confirms AI models run without runtime errors and correctly update database state.                    |
| External<br>Service<br>Mocking<br>Library | Libraries used to simulate responses from external APIs (e.g., Stripe, JazzCash, OpenAI).               | Payment Gateway Integration.<br>Translation/Sentiment API Calls.                                           | Ensures platform logic is robust against external service failures and unexpected responses.          |
| CI/CD<br>Pipeline<br>(GitHub<br>Actions)  | Automated workflow.                                                                                     | Scalability and Maintainability checks.                                                                    | Guarantees continuous code quality and deployment readiness by failing the build on any test failure. |

# **Chapter 7**

## **Conclusion**

# 7 CONCLUSION AND FUTURE WORK

## 7.1 Conclusion

The development of Taskio successfully achieved its primary objective: to create an intelligent, accessible, and trustworthy freelance task marketplace tailored to the unique socio-economic landscape of Pakistan. By adopting Agile Methodology within an Iterative Model, the project team was able to translate high-level requirements into a robust, functional, and modular application.

The core success of Taskio lies in its integration of novel Artificial Intelligence (AI) features, which directly address the systemic failures of existing platforms, namely inefficient resource allocation and a lack of transactional transparency. Key functional requirements successfully implemented include:

- **AI-Powered Intelligent Matching:** The hybrid algorithm was successfully implemented to analyze skill relevance, historical performance, and sentiment analysis scores, enabling the system to recommend the most qualified workers for a task.
- **Dynamic Pricing Model:** The AI-based regression model provides clients with an optimal price range during task posting, thereby promoting fair compensation and reducing market friction.
- **Core Marketplace Logic:** All essential transactional and security features, including secure User Registration and Authentication, the Task Management lifecycle, and Payment Integration using local APIs (JazzCash/EasyPaisha), were successfully deployed.

The decoupled, microservice-oriented architecture ensures that the system is scalable and maintains high performance by offloading resource-intensive AI computations to asynchronous Celery workers. Taskio stands as a comprehensive, technically validated solution ready to bridge the employment and task outsourcing gap in the local digital economy.

## 7.2 Future Work

While the current version of Taskio meets all defined requirements, continued development will focus on enhancing user convenience, security, and the sophistication of the AI Engine.

1. **Automated Scheduling and Availability Forecasting:** The platform's existing capability to manage worker schedules should be fully realized by implementing a robust system for availability forecasting. This feature would use time-series analysis to predict worker capacity and automatically allocate tasks, minimizing delays and improving service reliability.
2. **Dedicated Mobile Application Development:** To maximize accessibility for the local workforce, particularly non-technically skilled individuals, the system should move beyond the current responsive web design to develop native Android and iOS mobile applications.
3. **Advanced Identity Verification and Security:** The current identity verification process should be enhanced. This involves fully implementing the OCR/FaceNet Component to automate CNIC image validation, creating a significantly stronger trust mechanism against fraud and ensuring profile integrity.
4. **Multilingual and Voice Assistance Support:** The platform's Translation Module should be expanded to include comprehensive, native Urdu language support in the user interface and chat functionality. This will eliminate language barriers and broaden the platform's reach across the country.

## 8 REFERENCES

The final report's references are formatted using the IEEE standard and were compiled using EndNote, ensuring that all academic claims, implementation methodologies, and external context are fully supported by credible sources.

1. "Design Optimization of Data Pipelines in Gig Economy Platforms: Improving Data Processing Efficiency," *IEEE Conference Publication*, 2025. [link text](#)
2. D. Noever and F. McKee, "Can AI Freelancers Compete? Benchmarking Earnings, Reliability, and Task Success at Scale," *arXiv preprint arXiv:2505.13511*, 2025. [link text](#)
3. M. K. et al., "Skill matching at scale: freelancer-project alignment for efficient multilingual candidate retrieval," *CEUR Workshop Proceedings*, 2024. [link text](#)
4. Sahnoun and Elhadjamor, "Enhanced Freelance Matching: Integrated Data Analysis and Machine Learning Techniques," *Journal of Computing Theories and Applications*, vol. 1, no. 4, pp. 507-517, May 2024. [link text](#)
5. S. Naseem et al., "Survey on Sentiment Analysis of User Reviews," *2021 International Conference on Innovative Computing (ICIC)*, 2021. [link text](#)
6. "[Anonymous Authors]," "Predicting Stock Market Price Movement using Machine Learning Techniques," *IEEE Conference Publication*, 2022. [link text](#)
7. A. Sivaprakasam, M. Jazil, and P. S., "Revolutionizing the Gig Workforce: Technological Shifts and Emerging Opportunities," *2025 6th International Conference on Mobile Computing and Sustainable Informatics (ICMCSI)*, 2025. [link text](#)
8. J. R. et al., "Performance Analysis of Microservices Architecture using Django REST Framework," *International Journal of Computer Science and Network Security*, vol. 20, no. 8, pp. 112-118, 2020. [link text](#)
9. M. A. et al., "Implementing Real-Time Bidirectional Communication with Django Channels," *Journal of Network and Computer Applications*, vol. 15, no. 3, pp. 45-52, 2022. [link text](#)

10. M. K. et al., "A Comprehensive Study on User Authentication Mechanisms in Python Web Frameworks," *IEEE Security & Privacy*, vol. 18, no. 4, pp. 60-68, 2021. [link text](#)
11. A. M. H. et al., "A Deep Learning Approach for Price Recommendation in Digital Service Marketplaces," *2023 International Conference on Data Science and Machine Learning (ICDSML)*, 2023. [link text](#)
12. "[Anonymous Authors]," "Developing the Maturity Model for Gig Economy Business Processes," *IEEE Conference Publication*, 2019. [link text](#)
13. J. R. et al., "An intelligent chatbot system for customer support and task management in online platforms," *Journal of Intelligent Systems*, vol. 12, no. 2, pp. 88-95, 2022. [link text](#)
14. A. Sivaprakasam, M. Jazil, and P. S., "Automating Deployment in Web Applications Using CI/CD Pipelines with Docker and Kubernetes," *2024 International Conference on Software Engineering Practices (ICSEP)*, 2024. [link text](#)
15. M. K. et al., "Optimizing Resource Utilization using Containerization Technologies for Large-Scale Web Applications," *IEEE Transactions on Cloud Computing*, vol. 10, no. 1, pp. 45-55, 2023. [link text](#)
16. L. M. S. Lau et al., "Use Of VSP Resource Rooms to Support Group Work in a Learning Environment," *ACM 99*, pp. 2, 1999.
17. H. Nakanishi et al., "FreeWalk: Supporting Casual Meetings in a Network," *ACM CSCW*, pp. 308-314, 1996. [link text](#)
18. A. Behforooz and F. J. Hudson, *Software Engineering Fundamentals*. Oxford University Press, 1996, ch. 8, pp. 255-235.